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STATISTICAL AND MACHINE LEARNING APPROACHES EXPLORING BRAIN STRUCTURAL CHANGES IN ALZHEIMER'S DISEASE PROGRESSION: A LONGITUDINAL STUDY

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DECLARATION

We hereby declare that this project is the product of our own research and that none of it has been presented for credit toward any degree at this or any other university. All citations in this paper have been properly acknowledged.

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This is to certify that the project was undertaken through the candidates' independent effort and supervised in line with the University of Ghana's approved supervision guidelines.

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DEDICATION

We dedicate this research to the Almighty God, whose grace, strength, and guidance sustained us throughout this journey. To our families, friends, and loved ones, we extend our deepest gratitude for your unwavering support, constant encouragement, and immeasurable love. Your belief in us, your sacrifices, and your presence, whether near or far, gave us the motivation to persevere through every challenge. This work reflects your support, and we are forever grateful.

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ABSTRACT

Alzheimer's disease (AD) is one of the most significant contributors to cognitive decline and neurodegeneration in older adults. Timely intervention and management rely heavily on the early prediction of dementia progression. This research employed both statistical and machine learning methods to investigate the relationship between changes in brain structure and the progression of dementia, utilizing data from the Open Access Series of Imaging Studies (OASIS) longitudinal dataset.

Preliminary descriptive analyses were performed to examine the distribution of dementia severity across different groups. A one-way Repeated Measures MANOVA was used to evaluate the effects of group and time on cognitive and structural factors. Next, Principal Component Analysis (PCA) was applied to reduce dimensionality and identify key components that account for the most variance in the dataset. To address class imbalance related to dementia severity, the Synthetic Minority Oversampling Technique (SMOTE) was applied before training the predictive models. Five supervised machine learning methods—Gradient Boosting, Random Forest, Support Vector Machine, Neural Network, and Naïve Bayes—were developed and assessed using cross-validation and independent test datasets. The models' performance was evaluated with metrics such as accuracy, precision, recall, F1 score, and other relevant indicators.

Among all models, the Random Forest algorithm showed the highest predictive accuracy, effectively forecasting dementia progression and classifying subjects by severity. Age, MMSE, eTIV, and nWBV emerged as the key indicators of cognitive decline. The findings confirm the potential of machine learning to improve dementia diagnosis and progression modelling, while also supporting the use of neuroimaging and cognitive metrics for early detection.

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ABBREVIATION

AD	.....	Alzheimer's Disease
AdaBoost	.....	Adaptive Boosting
ADNI	.....	Alzheimer's Disease Neuroimaging Initiative
AUC	.....	Area Under Curve
BOLD	.....	Blood Oxygenation Level Dependent
BIRN	.....	Biomedical Informatics Research Network
CDR	.....	Clinical Dementia Rating
CDR-SOB	.....	Clinical Dementia Rating Sum of Boxes
CNN	.....	Convolutional Neural Network
CRT	.....	Cognitive Reserve
CSF	.....	Cerebrospinal Fluid
CV	.....	Cross-Validation
DMN	.....	Default Mode Network
DMTs	.....	Disease Modifying Therapies
F1	.....	F1 score (Harmonic mean of precision and recall)
FAQ	.....	Functional Activities Questionnaire
fMRI	.....	Functional Magnetic Resonance Imaging
FN	.....	False Negatives
FP	.....	False Positives
FPR	.....	False Positive Rate
GDS	.....	Global Deterioration Scale
HHMI	.....	Howard Hughes Medical Institute
IQ	.....	Intelligence Quotient
LMICs	.....	Low- and Middle-Income Countries
MANOVA	.....	Multivariate Analysis of Variance
MCI	.....	Mild Cognitive Impairment
MLP	.....	Multi-layer Perceptron
MMSE	.....	Mini-Mental State Examination
MRI	.....	Magnetic Resonance Imaging
NC	.....	Normal Control

NFT	.....	Neurofibrillary Tangles
NRG	.....	Neuroinformatics Research Group
OASIS	.....	Open Access Series of Imaging Studies
PCA	.....	Principal Component Analysis
RF	.....	Random Forest
ROC	.....	Receiver Operating Characteristic
SMOTE	.....	Synthetic Minority Over-sampling Technique
sMRI	.....	Structural Magnetic Resonance Imaging
SSA	.....	Sub-Saharan Africa
SVM	.....	Support Vector Machine
SVF	.....	Semantic Verbal Fluency
TN	.....	True Negatives
TP	.....	True Positives
TPR	.....	True Positive Rate
WHO	.....	World Health Organization

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CHAPTER ONE
Introduction

1.0 Overview

This chapter discusses how Alzheimer's disease (AD) results in significant changes in the brain's structure and function. It examines the effect of neurodegeneration on the decline in the brain's proper functioning and highlights the importance of tracking these changes as they unfold. It further assesses the role of Magnetic Resonance Imaging (MRI), a brain imaging technique, in detecting neurodegeneration and studying the structural changes in the brain associated with Alzheimer's disease.

1.1 Background

Alzheimer's disease (AD), a neurodegenerative disorder that impairs cognition, memory, and behavior, has shown an alarming rise in the global public health sector. Alzheimer's disease is proven to be the primary cause of dementia globally. It is increasingly becoming one of the most devastating, troubling, and lethal diseases of the century, and the incidence of dementia is predicted to triple by mid-century (Scheltens et al., 2021). Despite considerable efforts to attain effective treatment for Alzheimer's disease therapy, drug development has not achieved therapeutic success. In 2023, the World Health Organization (WHO) reported that over 57 million people have dementia worldwide, and Alzheimer's disease accounts for approximately 60–70% of these cases, with over 60% living in low- and middle-income countries (LMICs) (World Health Organization, 2023). Interestingly, the most recent data shows that dementia increases by approximately 10 million new cases annually. This is chiefly due to the rise in life expectancy and global population growth.

In North America, the prevalence of dementia is connected chiefly with the aging population. Approximately 6–10% of persons 65 years and older have dementia, with Alzheimer's disease (AD) contributing to about two-thirds of these cases. Alzheimer's Association

(2024) reports that approximately 7 million Americans are living with Alzheimer's, and it is projected to rise to nearly 13 million by 2025, provided no crucial breakthrough to preventing and treating Alzheimer-related diseases is made. Not surprisingly, Alzheimer's disease was the fifth leading cause of death among Americans aged 65 years and above in 2021.

Uwishema et al. (2024) report that Alzheimer's disease poses an existential and escalating public health threat across Africa, affecting approximately 2.67 million people, with a greater percentage living in Sub-Saharan Africa. African studies on Alzheimer's conducted in 2012 reveal that the incidence of Alzheimer's disease in Africa is 2.4%, predicted to rise drastically by mid-century, primarily driven by the aging population, increasing life expectancy, and significant risk factors, including low education, hypertension, smoking, and obesity (George-Carey et al., 2012).

In 2015, approximately 2.13 million people in Sub-Saharan Africa (SSA) were living with dementia. The number is expected to rise to about 3.48 million in 2030 and 7.62 million in 2050, respectively. A larger percentage of the predicted increase is likely to come from the Central and Eastern regions of the SSA, driven primarily by the rising aging population. In 2015, over 367,000 new cases of dementia were recorded (Guerchet et al., 2017).

Over the past few years, the incorporation of neuroimaging techniques has energized efforts to examine the structural brain changes associated with Alzheimer's. Magnetic Resonance Imaging (MRI) provides neurologists and researchers with detailed and non-invasive images of the periodic structural changes in the brain associated with Alzheimer's disease that may have gone unnoticed by human observers, for instance, the accumulation of β-amyloid plaques and neurofibrillary tangles (NFT), leading to neuronal damage and brain atrophy. A longitudinal MRI study would enable the analytic understanding of dynamic changes in the brain over time, including neurodevelopment and disease progression, making it essential for understanding cognitive and behavioral changes.

This research aims to analyze longitudinal MRI data to explore the brain's structural changes in individuals diagnosed with Alzheimer's. With the insights this study aims to reveal, researchers and neuroscientists will enhance their understanding of the disease's progression and develop resources for early detection and monitoring. This would align with ongoing initiatives to employ therapeutic strategies for improved healthcare outcomes in Alzheimer's disease.

1.2 Problem Statement

Alzheimer's disease continues to pose a significant challenge to the global healthcare systems, neuroscientists, and researchers. This is primarily due to the progressive nature of the disease and the current lack of a definitive cure, despite ongoing efforts to develop one (World Health Organization, 2025).

Alternative approaches have been sought to analyze and predict the onset and prevalence of Alzheimer's disease. A study by Jo et al. (2019) employed a deep learning approach, along with a combination of deep learning and traditional machine learning, for classification. The findings of their study resulted in about 98.8% precision for Alzheimer's disease classification and an 83.7% efficiency for predicting mild cognitive impairment (MCI) conversion to Alzheimer's disease. Similarly, a combination of multimodal neuroimaging and fluid biomarkers is expected to yield better classification performance.

Lee et al. (2019) employed a deep learning approach to monitor Alzheimer's disease progression by integrating a framework that combines cross-sectional brain imaging, longitudinal cerebrospinal fluid (CSF) biomarkers, and cognitive performance biomarkers. In addition, the outlined approach embedded multi-domain data to enhance the prediction accuracy. A prediction precision of 75% was achieved using a single data modality. Nevertheless, when longitudinal multi-domain data were employed, the prediction accuracy improved to 81%, resulting in a six-percentage-point enhancement.

Liu et al. (2014) employed deep learning techniques to propose a method for the timely diagnosis of Alzheimer's disease, which concurrently reduces dimensionality and data fusion to preserve the unified effort between data modalities. When evaluated against a support vector machine (SVM) classifier, the system was observed to achieve a performance advantage in terms of overall accuracy and specificity

Despite advancements in deep learning techniques for predicting Alzheimer's, significant gaps still exist. The use of longitudinal structural MRI data to model disease progression is limited; reliance on multimodal datasets that are not widely accessible persists, and there is a lack of effective integration between dimensionality reduction methods and classification models for dementia severity, which increases the risk of overfitting. This research addresses these problems by employing PCA-based feature extraction along with various machine learning algorithms to predict and classify the progression of Alzheimer's disease based on a purely structural and demographic longitudinal dataset.

1.3 Research Objectives

The primary objective of this study is to utilize longitudinal MRI data to investigate and examine the progressive brain structural changes and cognitive decline associated with Alzheimer's disease.

The specific objectives subsume;

1.3.1. To classify and predict subjects on dementia severity.

1.3.2. To investigate whether there is a connection between dementia status and combined observable variations in brain structure and cognitive functions.

1.4 Significance of the Study

This study provides valuable insights to the global health community by delivering scientific and medical information essential for understanding the gradual effects of Alzheimer's disease on brain anatomy throughout its stages. Such knowledge facilitates early detection, leading to prompt intervention, enhanced patient care, and accelerated development of therapeutic strategies.

The proliferating aging of the population, associated with the rising number of people with Alzheimer's disease, and the urgent need to document prognosis, trigger the study of the progressive structural brain changes. Longitudinal research is crucial for elucidating the nature of AD. Ultimately, it enables researchers, neurologists, and clinicians to distinguish clearly between the stages and subtypes of Alzheimer's disease. Applying machine learning algorithms to longitudinal MRI data enhances the timely identification of high-risk individuals for dementia by addressing potential limitations of human assessment, such as uncovering hidden and complex patterns in the data.

This research aims to benefit as many users as possible. Based on these findings, healthcare systems could enhance the accuracy of AD detection, leading to targeted therapy and improved patient outcomes. MCI subjects stand to gain from the findings of this research, which can help them better protect themselves from progressing into AD. Similarly, persons diagnosed with AD, as well as caregivers, could benefit from the earlier detection and mitigation strategies by the healthcare systems to decelerate the progression of the disease, thereby reducing the risk of dementia diagnosis. Neuroscientists and researchers could also benefit from the findings of this study, as they provide valuable insights into the relevance of biomarkers (such as MRI scans) in AD progression research, potentially leading to further studies.

1.5 Brief methodology

This study would employ secondary data analysis. The data used in this research were procured from a database available at `https://www.kaggle.com/datasets/jboysen/mri-and-alzheimers/data`. The Open Access Imaging Studies (OASIS) initiative aims to provide the scientific community with a freely available MRI dataset to enhance future discoveries in basic and clinical neuroscience. Included in the OASIS dataset are: Cross-sectional MRI Data in Young, Middle Aged, Nondemented and Demented Older Adults, and Longitudinal MRI Data in Nondemented and Demented Older Adults. This study focused on longitudinal MRI Data in Non-demented and demented older adults. The set constituted an initial sample size of 150 subjects aged 60 to 96. Multiple scans, with an interval of at least one year, were performed on these subjects, totaling 373 imaging sessions. From the initial 150 subjects, 72 and 64 remained non-demented and demented, respectively, throughout the imaging sessions. Fourteen subjects were, in their initial scans, labeled as non-demented, but later scans characterized them as demented.

The statistical approaches and machine learning methods employed in this research include Multivariate Analysis of Variance (MANOVA), Principal Component Analysis (PCA), and other classification methods that may evolve as the study progresses. These data analysis techniques would enable the concurrent assessment of differences in MRI variables between demented and non-demented subjects, allowing the transformation of a large set of variables into smaller sets without omitting the substantive information in the original data.

1.6 Tools for Data Analysis

The primary tool used in this study will be the R statistical software. The selection of R for this study is based on its extensive range of powerful packages for statistical modeling, data visualization, and machine learning algorithm development, all of which are suitable for analyzing the OASIS MRI longitudinal data. This enables the acquisition of insights into data trends and patterns, allowing for the making of appropriate statistical inferences.

1.7 Chapter Organization

This study is structured into five chapters, each serving a targeted purpose in exploring the structural brain changes associated with Alzheimer's disease.

Chapter one introduces the topic by providing a brief overview of Alzheimer's disease, highlighting its significance as a global health issue and how the brain anatomy responds

to the progression of the disease. It also presents the problem statement, research objectives, methodology, and the tool for analyzing the data.

Chapter two reviews existing relevant literature on Alzheimer's disease, examining previous research findings, the theoretical framework, and empirical inspection that inform the relationship between Alzheimer's disease and the structural brain changes. This review will outline the gaps this study aims to address.

Chapter three details the methodology used in this research. It will explain the data source and acquisition method. Additionally, it will explain the variables in the OASIS longitudinal MRI data and the techniques utilized for data analysis.

Chapter four analyzes and presents the MRI data acquired. Descriptive statistics will initially summarize the data and then make inferences. Regression analysis and hypothesis testing will be performed to estimate the relationship between variables in the data.

Chapter five summarizes the key findings of the research, drawing conclusions based on the data analysis. Practical recommendations will also be drawn for public health and policy-making. It will provide suggestions for future studies that aim to address the setbacks encountered during this study.

CHAPTER TWO
Literature Review

2.0 Overview

Alzheimer's disease is primarily referred to as a brain defect that affects memory and thinking skills, ultimately causing neurocognitive disorders. On average, AD commences after 60 and may persist for 8 to 12 years. There are two major types of Alzheimer's disease: early-onset Alzheimer's disease and late-onset Alzheimer's disease. Early-onset Alzheimer's disease commences in advance of age 65 and is considered to be chiefly hereditary, although not phenotypically expressed (Ridge et al., 2013). Late-onset Alzheimer's disease begins post age 65 and is the most common form of Alzheimer's, contributing to approximately 90–95% of the gross Alzheimer's cases globally. While both types progress, early-onset AD progresses rapidly and exhibits swift cognitive deterioration.

2.1 History of Alzheimer's Disease

Auguste was born to the Kassel family in Germany in the mid-19th century. At age 23, she married Karl Deter and later cohabited with him in Frankfurt. Twenty-eight years later, Auguste's life became increasingly chaotic, full of sudden accusations, hallucinations, and alarming neighbors with a terrifying scream. Her mental health declined swiftly, and she began to mistrust the actions of unfamiliar individuals without cause.

Owing to the rapidly deteriorating condition of Auguste's cognition, Karl, his husband, transported her to a mental hospital for medical attention. A medical practitioner named Dr. Alois Alzheimer closely monitored her condition. Following a thorough review of the patient's history, Alzheimer determined that the case was a form of "presenile" dementia and requested to own the patient's clinical history and brain after her demise (Yang et al., 2016). Soon after Dr. Alois Alzheimer's exit from the mental hospital, Mrs. Auguste Deter's condition worsened significantly, causing her inability to perform daily tasks, including eating. On April 8, 1906, four years after Alzheimer's departure, Mrs. Auguste

passed away, following her prolonged illness related to septicemia and pneumonia due to blood poisoning and pressure injuries, as revealed by the initial post-mortem examinations (Aminoff, 2014). The medical history, together with the brain of the deceased, was handed over to Dr. Alzheimer upon his earlier request.

Yang et al. (2016) report that Alzheimer conducted detailed research on the brain to track the connection between the clinical records and the symptoms that unfolded. He realized an atrophy in the cerebral cortex, responsible for disrupting the brain's higher-level functions, including thinking, language, and memory. The accumulation of plaques around the neurons and the development of tangles in the nerve fibers were also observed, which can be characterized as early-onset Alzheimer's disease in recent times. These findings were presented at a medical conference in April 1906 at Tubingen by Alois Alzheimer himself, although they initially failed to attract noticeable attention from distinguished scholars. In 1910, Dr. Emil Kraepelin officially coined the eponym "Alzheimer's disease" in the eighth revised edition of his Clinical Psychiatry book, acknowledging Dr. Alois Alzheimer's research findings in Auguste Deter's case (Ramirez-Bermudez, 2012; Yang et al., 2016).

Before the mid-20th century, Alzheimer's disease was considered to be infrequent, reserved for individuals before age 65 years, and termed as presenile dementia. Senile dementia was ascribed to dementia syndromes that develop after age 65 years and are thought of as part of normal aging. However, researchers in the late 1960s discovered that Alzheimer's disease ignores the age restriction placed on it, as the same neuropathologic lesions that distinguish presenile dementia were seen to be present in individuals above the age of 65 years (Katzman, 1976; Villain and Michalon, 2024).

This resulted in the reclassification of Alzheimer's disease as the most typical cause of dementia in individuals, regardless of age group (Katzman, 1976; Villain and Michalon, 2024).

The late 20th and early 21st centuries marked a period of rapid development in inspecting the neurobiological framework, as well as the diagnostic and therapeutic focus of the disease. Beta-amyloid, Aβ, was discovered as the main component of amyloid plaques in Alzheimer's disease in 1984 (Huynh and Holtzman, 2018). Kosik et al. (1986) report that tau-protein, τ, is a primary antigenic component of neurofibrillary tangles, NFTs, which are associated with the disease (Kosik et al., 1986). The unveiling of neuroimaging techniques through the establishment of the Alzheimer's Disease Neuroimaging Initiative (ADNI) in 2004 aimed to identify the relationships among the clinical, cognitive, and genetic features associated with the disease's spectrum (Weiner et al., 2010). These discoveries were instrumental and remain the most promising standardized medical protocols for the timely detection and disease progression, as there is no current definitive

cure for Alzheimer's disease.

Initial statistical evaluations of the disease focused on descriptive statistics, correlational studies, and basic regression methods to measure its prevalence and connect cognitive assessment results, clinical demographics, and brain-derived characteristics to neuropathological findings. The earliest comprehensive statistical analysis on Alzheimer's disease seems to be a study from 1993 conducted by Azari et al., which utilized positron emission tomography (PET) data alongside sophisticated statistical methods to explore discriminant functions that classify AD patients from controls, achieving an 87% classification accuracy (Azari et al., 1993). Their cross-sectional study was limited by a smaller sample size, which challenged the generalizability of the results, and a suggestive result.

Foundational works on longitudinal cohort studies, alongside the implementation of neuroimaging data, closely followed those on cross-sectional studies in the late 20th century through to the early 21st century, focusing on growth and survival analysis in modeling rates of cognitive decline and disease progression. The work of McArdle et al. (2005) was central to this. In this study, new statistical methodologies were employed, utilizing longitudinal data to estimate survival and quantify changes in the early prediction of the disease, yielding optimal predictors of impending dementia (McArdle et al., 2005). The limited sample size again affects the precision of the estimates for the shared-parameter models.

Over the past twenty years, studies on Alzheimer's disease have increasingly focused on statistical learning and machine learning methods, such as Random Forest, SVM, Gradient Boosting, Neural Networks, and ensemble approaches, which deliver improved predictive accuracy for both identifying the disease and forecasting its progression.

2.2 Theoretical framework

2.2.1 Stages of Alzheimer's Disease

Alzheimer's disease evolves progressively, with the pathophysiological changes commencing many years before the clinical symptomology develops along a spectrum. This earliest phase of the disease, before the clinical manifestation, is termed preclinical Alzheimer's disease. According to the Alzheimer's Association, an Alzheimer's patient's life spans, on average, four to eight years after diagnosis, although one can live as long as 20 years. Despite the patient-specific, progressive, non-linear, and multifaceted nature of AD, various staging models have been established to classify the disease into discrete, distinct, and defined clinical stages, which are critical for developing robust disease-modifying therapies (DMTs) (Alzheimer's Association, 2024; Butler Hospital, nd).

Early Stage (Mild) Alzheimer's

The early stage is the first clinical phase of AD and may precede formal clinical diagnosis. This stage is characterized by pronounced memory deficit. At this stage, the patient experiences reduced social functions, may encounter difficulty with language, and undergoes changes in behavior and mood, which typically last approximately three to four years. However, the individual performs uninterrupted daily tasks independently. Whitwell et al. (2007) report that atrophy in the hippocampus becomes visible in this stage through brain imaging findings, resulting in a volume depletion of the brain region responsible for memory and learning (Whitwell et al., 2007). During this stage, symptoms may only be detectable to the patient and their closest individuals. According to Butler Hospital, obtaining a diagnosis in this stage enhances improved planning and management throughout the progression of the disease (Butler Hospital, nd).

Middle Stage (Moderate) Alzheimer's

The middle stage is generally the longest in duration of AD and may persist for years. During this stage, the Alzheimer's patient grows more confused and forgetful. Patients in this stage are unable to perform social functions independently and may require assistance with basic daily tasks. The Alzheimer's Association reports that, with the growing demand for more intensive care for Alzheimer's patients in this stage, caregivers might explore options like temporary care services to provide a safe and supportive environment for the patient. This allows caregivers a brief respite while ensuring the patient receives the necessary care (Butler Hospital, nd).

According to a report by Mayo Clinic (2025), the affected individual may wander, perhaps searching for a more recognizable environment, as disorientation in time, place, and unfamiliar environment sets in (Mayo Clinic, 2025). The brain structure undergoes significant deterioration at this stage as the disease progresses throughout the brain. Atrophy of the brain continues progressively, outstretching to the parietal lobe, as well as thinning of the cerebral cortex (Vernooij and van Buchem, 2020).

Late Stage (Severe) Alzheimer's

A significant cognitive deficit characterizes the final stage, and the patient may require intensive around-the-clock care. The AD-diagnosed individual in this stage loses physical ability, hence becomes entirely dependent on caregivers to be able to carry out basic daily activities such as walking and maintaining personal hygiene. The patient becomes vulnerable to infections, predominantly aspiration pneumonia, due to difficulty in swallowing and reduced ambulation. Complete disorientation of the environment, as well as a decline in the patient's conventional communication, is welcomed in this stage. The brain undergoes volumetric atrophy and extensive damage, disrupting approximately its entire function (Butler Hospital, nd; Alzheimer's Association, 2024).

2.2.2 Cognitive Theories and Models of Decline

Theoretical models of cognitive decline focused on understanding and appreciating how Alzheimer's disease alters thought patterns and behavior have been of crucial concern. These models seek to underscore the interplay between the biological, psychological, and social factors of cognitive decline, aiding in explaining the relationship between brain pathology and AD. In essence, these frameworks enhance our appreciation of neuroimaging data and forecast the disease's trajectory. These theories include Cognitive Reserve Theory (CRT), Disconnection Theory, and models connecting cognition with brain structural changes.

- **Cognitive Reserve Theory**

  The Cognitive Reserve Theory posits that the brain makes an effort to manage brain damage before clinical manifestation (Stern, 2013). This model attempts to explain the individual's varying capacities in surviving brain pathology. The individual variability in cognitive reserve can be attributed to genetic differences or life experiences, such as educational attainment, intellectually demanding occupational experiences, leisure activities, and other social engagements (Stern, 2013; Tucker and Stern, 2014). Stern et al. (1994) and Tucker and Stern (2014) report that these variables tend to be positively related to cognitive reserve and better clinical outcomes; thus, higher levels of education, occupational attainment, and intelligence quotient (IQ) are accurate forecasters of high cognitive reserves. The theory presupposes that the reserve utilizes alternate brain networks and strategies to maintain performance amidst brain damage. Hence, there is no explicit relationship between the degree of brain pathology and observed cognitive functional deficits (Stern, 2013; Stern et al., 1994; Tucker and Stern, 2014).

- **Disconnection Theory**

  The disconnection theory proposes that cognitive decline associated with Alzheimer's disease results from the inability of the brain's communication networks to interconnect optimally, rather than solely from focal neuronal loss (Bokde et al., 2009). Sanz-Arigita et al. (2010) report that the unusual reorganization of unimpaired functional connectivity networks results in cognitive decline in AD patients. The progressive demyelination of the white matter and the damage to the synaptic connections disrupt the effective transmission of information across the brain regions, which plays a crucial role in the pathogenesis and diagnosis of the disease (Sachdev et al., 2013). According to Jones et al. (2011), reduced connectivity in brain networks, such as the default mode network (DMN), accounts for the low activity observed during internally focused mental states.

- **Models Linking Cognition with Brain Structure**

  Various models have been proposed to explain how brain structure supports cognitive and functional processes. These models link cognitive decline to the brain structural changes.

  The **Sequential Model of Decline** posits that Alzheimer's disease follows a consistent pattern of cognitive deterioration; thus, several staging systems describe the disease's predictable progression. Henneges et al. (2016) identify the sequence of cognitive decline in AD patients as starting in impairment in recall, followed by impairment in orientation in time, attention, and concentration, and then in orientation in place, language, drawing, and registration (Henneges et al., 2016).

  The **Atrophy-Cognition Model** suggests that the progressive neurodegeneration in distinct brain regions of an AD patient corresponds to a gradual decline in cognition. Thus, as the disease progresses, measurable shrinkage of notable brain regions, such as hippocampal atrophy, results in equivalent cognitive symptoms.

  The **Biomarker-Cognition Model** incorporates recent diverse techniques in understanding how diagnostic indicators of Alzheimer's disease associate with observable cognitive symptoms, subsuming episodic memory loss, among others. Jack et al. (2010) propose a model associating the stages of the disease with AD biomarkers: beta-amyloid (Aβ) biomarkers become abnormal first, followed by neurodegenerative biomarkers and neuropsychological manifestations that connect with clinical symptom severity (Jack et al., 2010).

2.2.3 MRI as a Tool for Studying Neurodegeneration

Magnetic Resonance Imaging (MRI) has been identified as a high-powered tool for studying neurodegeneration, offering detailed structural and functional non-invasive images of the brain in investigating the gradual alterations of the brain as the disease progresses. Huang and Zhang (2023) report that MRI has been extensively utilized in both animal research and large-scale clinical investigations, providing a linkage between micro- and macro-level analysis and promoting bench-to-bedside translational research due to its non-invasive and non-radioactive nature. In Alzheimer's research, the two basic types of MRI used are structural MRI (sMRI) and functional MRI (fMRI).

Structural Magnetic Resonance Imaging (sMRI)

Structural magnetic resonance imaging is a neuroimaging tool used to visualize and quantify the brain morphometry; it produces high-resolution, detailed 3D images of the brain and detects gray matter atrophy, cortical thinning, and ventricular enlargement linked to neuronal and synaptic losses that occur on a microscopic level in AD (Vemuri and Jack,

2010). Volumetric variations of the distinct brain regions associated with AD progression are also measured by sMRI scans, facilitating comparison among AD patients as well as the disease's stages.

Functional Magnetic Resonance Imaging (fMRI)

Functional magnetic resonance imaging, on the other hand, is a non-invasive tool used to visualize brain activity and network connections. Buxton (2013) reports that fMRI is used to detect the dynamic patterns of activity in the active human brain. The fMRI method mainly relies on blood oxygenation level-dependent (BOLD) contrast, which is based on the contrasting magnetic properties of oxygen-rich and oxygen-poor blood (Buxton, 2013; Logothetis et al., 2001). Gupta et al. (2010) note that fMRI plays an increasingly important role in preoperative planning by identifying eloquent brain regions in relation to intracranial pathologies, as well as enhancing our understanding of the neural basis of significant psychiatric disorders, neurological conditions, and substance use disorders.

2.3 Empirical Framework

2.3.1 Brain Atrophy Patterns and Progression in Alzheimer's disease

A substantial amount of research has utilized neuroimaging techniques to explore the structural brain alterations associated with Alzheimer's disease (AD). These investigations have been conducted using both cross-sectional and longitudinal designs, which enable researchers to assess the current state of the disease (cross-sectional) and track its impact on biomarkers and brain anatomy over time as the disease progresses (longitudinal).

Cross-sectional studies of AD have revealed appreciable progressive brain deformation, including hippocampal shrinkage, cortical thinning, and ventricular enlargement, when compared to age-matched healthy controls. For instance, Apostolova et al. (2012) conducted a cross-sectional study of Alzheimer's disease on MRI data from 46 cognitively normal elderly individuals (NC), 33 participants with mild cognitive impairment (MCI), and 43 AD subjects to analyze the hippocampal and ventricular volumes using two innovative semi-automated segmentation methods accompanied by the radial distance mapping techniques. In their study, multiple regression was employed to evaluate the impact of age and diagnosis on hippocampal and ventricular volumes, as well as radial distance. Their findings suggested that the most pronounced hippocampal atrophy and ventricular enlargement were observed in the comparison between AD and NC. Subjects with MCI exhibited intermediate levels of both hippocampal atrophy and ventricular enlargement. Significant age-related effects on hippocampal volume and radial distance were identified

in the combined sample, as well as in the NC and AD groups separately (Apostolova et al., 2012).

Jack et al. (2000) analyzed cross-sectional data on 129 subjects from the Mayo Clinic AD Research Center who underwent MRI examinations of the head, recording the annualized percentage change in hippocampal volume. Findings from Jack et al. (2000) highlight that the annualized rates of hippocampal volume reduction for the three initial clinical groups showed a progressive decline in the following order: AD > MCI > NC. Among the control and MCI groups, those who experienced a decline had a significantly higher rate of volume loss compared to those who remained clinically stable. The average annualized rates of hippocampal atrophy by follow-up clinical group were as follows: control-stable (1.73 percent), control-decliner (2.81 percent), MCI-stable (2.55 percent), MCI-decliner (3.69 percent), and AD (3.5 percent). The strength of cross-sectional research lies in its efficiency and reduced expense; however, its drawback is the inability to track changes over time, which can lead to ambiguity regarding whether the observed differences indicate actual disease progression or existing anatomical variability.

Longitudinal studies of MRI research have offered an important understanding of brain atrophy patterns throughout the spectrum of Alzheimer's disease, as it tracks changes in the progressive deformation of the brain. Morra et al. (2009) undertook a longitudinal analysis of the 3D pattern of hippocampal degeneration over a period of time in 490 participants who underwent two brain MRI scans spaced a year apart (totaling 980 scans). In their work, Morra et al. (2009) evaluated the initial and one-year follow-up scans for 97 individuals with Alzheimer's disease (49 males and 48 females), 148 healthy control participants (75 males and 73 females), and 245 individuals with mild cognitive impairment (MCI; 160 males and 85 females). The Adaptive Boosting algorithm (AdaBoost) was employed to generate 3D models of the hippocampal surface from all 980 scans. Results revealed an increased rate of hippocampal volumetric loss with the severity of diagnosis (normal = 0.66%/year; MCI = 3.12%/year; AD = 5.59%/year) and was associated with both initial and periodic changes in Mini-Mental State Examination (MMSE) scores, as well as global and sum-of-boxes Clinical Dementia Rating scale (CDR) scores. Additionally, surface-based statistical maps illustrated a distinct pattern of ongoing atrophy across all three diagnostic categories (Morra et al., 2009).

Similarly, using a longitudinal approach, Sluimer et al. (2008) investigated the brain of 65 patients with Alzheimer's disease (38 females, 27 males; aged 52 to 81 years), 45 patients with mild cognitive impairment (22 females, 23 males; aged 56 to 80 years), 27 individuals with subjective complaints (12 females, 15 males; aged 50 to 87 years), and 10 healthy controls (six females, four males; aged 53 to 80 years), with two MRI scans obtained at a mean interval of 1.8 years ± 0.7 (standard deviation) used in assessing the baseline brain volume and rates of whole-brain atrophy. Findings from Sluimer et al. (2008) indicate that

brain volume was minimal in AD patients, though not significantly different from MCI groups, subjective complaints, and healthy controls ($p > 0.38$). As expected, Sluimer et al. (2008) realized a greater whole-brain shrinkage in AD patients (-1.9% per year $\pm$ 0.9), followed by the MCI patients (-1.2% per year $\pm$ 0.9), with MCI patients exhibiting higher atrophy rates than those with subjective complaints (-0.7% per year $\pm$ 0.7) and controls (-0.5% per year $\pm$ 0.5). These provide valuable insights into disease progression and patterns of change, enabling researchers to identify environmental and genetic factors that increase AD risk and contribute to cognitive decline. Global surveys have revealed that these studies often face considerable methodological challenges (unadjusted heterogeneity in demographic factors such as education and age between subjects, attrition, selective survival, and neuropsychological assessment error) that compromise research validity.

2.3.2 Cognitive Decline Assessment

Considerable cognitive evaluation tools, such as the Mini-Mental State Examination (MMSE) and the Clinical Dementia Rating (CDR), are used to screen and classify the disease into its distinctive, non-overlapping stages and track the disease's progression. Earlier publications have demonstrated a substantial correlation between brain structural deformation and MMSE and CDR scores. Mouton et al. (1998) assessed a subset of 46 subjects on their performance on the Mini-Mental State Examination (MMSE). The report revealed that probable AD cases exhibited clinical signs of unexplained, severe, and progressive dementia (final MMSE score < 25; mean 6.7, SD 6.6), whereas the controls displayed no signs of neurological disease (final MMSE scores ≥ 25; mean 28.4, SD 1.5), indicating the presence of a strong correlation between cognitive function and regional brain volume (Mouton et al., 1998). Comparable results by Jack et al. (2002) showed that AD patients with reduced hippocampal volumes had lower scores on the MMSE ($r = 0.60$), suggesting a direct connection between atrophy in the medial temporal lobe and memory deficits. Grundman et al. (2002) conducted a correlational analysis between the MRI hippocampal volumes recorded at the start of the study and the subjects' current performance on neuropsychological assessments and clinical evaluations. The analysis suggested that larger hippocampal volume correlated with improved performance on memory tests, overall cognitive measures, and general clinical assessments. It also indicated that a lower hippocampal volume at baseline might be related to an increased risk of progressing to clinical AD. Grundman et al. (2002) concluded that MRI hippocampal volume could serve as a valuable indicator of disease severity in MCI individuals and may act as a prognostic measure for the future development of AD.

Statistical and machine learning models have been established to forecast and analyze cognitive assessment tools, including MMSE, CDR, and CDR Sum of Boxes (CDR-SOB) scores, based on initial MRI characteristics. Linz et al. (2017) employed regression models

on over 179 AD patients with degrees of dementia to predict MMSE and CDR scores using baseline Semantic Verbal Fluency (SVF) performance. The analysis from Linz et al. (2017) recorded a mean absolute error of 2.2 for the MMSE (with a range of 0–30) and 1.7 for the CDR-SOB (with a range of 0–18). The correlation between actual and predicted scores showed a Cohen's κ of 0.76 for MMSE and 0.52 for CDR-SOB (Linz et al., 2017). Sato et al. (2024) utilized baseline data from 944 participants in the Alzheimer's Disease Neuroimaging Initiative (ADNI) and 171 J-ADNI amyloid-positive participants, focusing on Mini-Mental State Examination (MMSE) and Functional Activities Questionnaire (FAQ) scores to establish a machine learning model for forecasting variations in CDR-SOB. The model by Sato et al. (2024) attained the necessary prediction accuracy (MAE $<$ 1.0) and demonstrated satisfactory predictive performance ($R^2 > 0.5$) for patients with mild cognitive impairment (MCI) regarding changes in CDR-SOB over periods of 18 months or longer. Moreover, Perneczky et al. (2006) formulated MMSE scores in relation to CDR stages. Cohen's measure of agreement demonstrated a significant agreement for mild ($\kappa = 0.62$), moderate ($\kappa = 0.69$), and severe dementia ($\kappa = 0.76$), justifying their proposition that MMSE could be used as a substitute measure for CDR staging.

Nonetheless, several gaps are apparent in the existing literature. The lack of integration of MRI data with longitudinal cognitive trajectories, inadequate comprehensive model validation, and the crucial question of how correlations are assessed raise concerns about the robustness and accuracy of these predictive models, as well as their clinical relevance.

2.3.3 Machine Learning Applications in Alzheimer's Research

The swift emergence of machine learning algorithms as a ground-breaking resource for examining neuroimaging data to address the complex diagnostic and forecasting challenges in Alzheimer's research has raised some alarms. The far-reaching application of machine learning techniques has been employed to classify AD patients from healthy controls automatically and to establish predictive models for forecasting the conversion of MCI to AD. Multiple machine learning algorithms are featured prominently in the existing literature. The Support Vector Machine (SVM) algorithm has been applied extensively on an MRI dataset for AD classification and regression tasks. Kruthika et al. (2018) utilized a support vector machine (SVM), accompanied by an sMRI data-derived anatomic brain features from 360 subjects processed by a FreeSurfer, to classify AD stages consisting of 130 standard controls, 150 mild cognitive impaired patients, and 80 AD patients, all sourced from Alzheimer's Disease Neuroimaging Initiative (ADNI). Publication by Yepes-Calderon et al. (2014) presents a straightforward approach for rapid diagnostic classification in clinical settings utilizing Support Vector Machines (SVM) and easily obtainable geometrical measurements, combined with cortical and sub-cortical brain parcellation. The classification model achieved classification success rates of up to 99.2% from a sin-

gle measurement using an extensive imaging dataset comprising more than 800 subjects from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database. Furthermore, the Random Forest (RF) is a widely used supervised machine learning algorithm that has been extensively applied in neuroimaging to manage complex, high-dimensional brain data in Alzheimer's classification. Song et al. (2021) tested the classification accuracies of random forest and other machine learning algorithms (Support Vector Machine (SVM), Multi-layer Perceptron (MLP), and Convolutional Neural Network (CNN)) with regional volumes of 2250 subjects from the ADNI database. The RF model achieved a classification accuracy of 90.2%, ensuring stability even when feature sets are limited. This suggests that RF forecasts AD more consistently with a reduced number of features (Song et al., 2021). Of particular relevance, the RF algorithm provides an accurate auto-calibrated framework for selecting a small set of jointly informative regions due to its high predictive accuracy and the use of variable importance measures (Genuer et al., 2010a; Fabris et al., 2018).

It is essential to acknowledge specific limitations that hinder the advantages of using machine learning algorithms compared to traditional statistical methods. The neglect of significant clinical and demographic factors is prevalent throughout the literature. This lack of integration restricts the model's generalizability and its application in real-world scenarios. Furthermore, employing a limited dataset leads to overfitting the model, which compromises its reliability.

2.4 Summary

This chapter has reviewed the historical and theoretical foundations, as well as the empirical studies connected to Alzheimer's disease. The literature heavily demonstrated how the disease progressively deteriorates the brain structure using the notable magnetic resonance imaging technique. Cross-sectional studies of Alzheimer's Disease (AD) have offered significant insights into understanding the disease's prevalence, risk factors, and its effects on brain structure. However, these studies often do not track disease progression, which limits their ability to establish causal relationships because they are based on one-time assessments. Longitudinal studies provide a more comprehensive understanding by allowing for the observation of the disease over time, thus facilitating the establishment of causal links.

The empirical framework highlights the connection between structural brain changes related to AD and cognitive assessments, such as the Mini-Mental State Examination and Clinical Dementia Rating scores. The substantial correlation documented in the literature supports the use of these clinical assessment tools as alternative biomarkers for gauging disease severity. The introduction of various machine learning algorithms for

automating AD classification has further enhanced diagnostic and prognostic capabilities related to the progression of AD.

The existing literature reveals significant methodological weaknesses in AD research, particularly the failure to incorporate clinical demographic variables in predicting AD classification models. Additionally, the use of relatively small sample sizes in previous studies restricts the applicability of their findings and limits clinical relevance. By addressing these issues, this study seeks to build upon the current research by utilizing a large, well-defined longitudinal dataset (OASIS) and applying robust machine learning algorithms to analyze changes in brain structure over time.

CHAPTER THREE
Methodology

3.0 Overview

The impact of Alzheimer's disease (AD) cannot be given disproportionate weight. The social, economic, and psychological effects of the disease are not only borne by individuals but also by their families, caregivers, society, and the broader global community. Despite the diagnostic and prognostic advancements in medical practice for AD prevention and treatment, the worldwide control of AD is still far from optimum, as drug development to cure AD has not achieved therapeutic success. Thus far, various strategies have been developed to reduce behavioral risk factors, such as maintaining a nutritious diet, enhancing social and cognitive skills, promoting engagement and awareness, ensuring adequate sleep, avoiding smoking, and practicing self-management. This chapter describes the statistical modeling strategy and machine learning algorithms used to analyze longitudinal brain structural changes in Alzheimer's disease, including the validation methods employed to confirm the reliability of the results.

3.1 Data Acquisition

The longitudinal magnetic resonance imaging (MRI) dataset from non-demented and demented older adults, made freely available to the scientific community by the Open Access Series of Imaging Studies (OASIS), is used for this study. This dataset comprises a longitudinal study of 150 individuals aged between 60 and 96. Each subject underwent at least two scans, with the scans spaced at least one year apart, resulting in a total of 373 imaging sessions. The OASIS collaborated with the Alzheimer's Disease Research Center at Washington University, Dr. Randy Buckner from the Howard Hughes Medical Institute (HHMI) at Harvard University, the Neuroinformatics Research Group (NRG) at the Washington University School of Medicine, and the Biomedical Informatics Research Network (BIRN) in making the dataset freely accessible to the neuroscience field. These organizations and prominent figures have dedicated research groups and resources for

further investigations.

The dataset comprises multiple categorical and numerical variables, grouped into the following categories:

- **MRI-derived features:** Estimated Total Intracranial Volume (eTIV), Normalized Whole Brain Volume (nWBV), and Atlas Scaling Factor (ASF).
- **Clinical Demographic Variables:** Subject ID, MRI ID, Group (Demented or Nondemented), Visit, MRI Delay, Gender (Male or Female), Hand (Left or Right), Age, Education (EDUC), and Socioeconomic Status (SES).
- **Cognitive Evaluation Tools:** Mini-Mental State Examination (MMSE) score, and Clinical Dementia Rating (CDR).

Table 3.1: Dataset Composition

Variable	Description	Values / Categories
Subject ID	A distinct identifier for each subject in the dataset.	Nominal (from OAS2_0001 to OAS2_0186).
MRI ID	The specific MRI session or scan identifier for each subject in the dataset.	Nominal (from OAS2_0001_MR1 to OAS2_0186_MR3).
Group	This indicates whether the subject is part of a control group, has Alzheimer's Disease (AD), or belongs to another specific group.	Demented, Nondemented, or Converted.
Visit	This indicates the number of times a subject was scanned to track the changes over time.	Discrete (1–5).
MR Delay	This represents the time (delay) between different MRI scans or sessions on a subject.	Continuous (0–2639).
M/F	This denotes the gender of each subject.	Male or female.
Hand	A specification on the dominant hand of the subject, though in this case, all subjects are described as right-handed.	R.
Age	This indicates the age of the subject at the time of the MRI scan.	Continuous (60–98).

Table 3.1 continued

Variable	Description	Values / Categories
EDUC	Represents the years of education or the level of education completed by the subject.	Discrete (6–23).
SES	A measure of a subject's economic and social position relative to others.	Ordinal (1–5).
MMSE	Mini-Mental State Examination score, a test to assess cognitive function.	Continuous (4–30).
CDR	Clinical Dementia Rating, a scale used to quantify the severity of symptoms of dementia.	Continuous (0–2).
eTIV	Estimated Total Intracranial Volume, referring to the size of the intracranial cavity based on MRI data.	Continuous (1106–2004).
nWBV	Normalized Whole Brain Volume, an MRI-derived measure that adjusts for total brain volume.	Continuous (0.644–0.837).
ASF	Atlas Scaling Factor, a metric used to normalize or standardize brain volumes to a standard atlas or template.	Continuous (0.876–1.587).

This data aids in comprehending the causes, catastrophic effects, and toll of Alzheimer's disease. These variables are examined to identify and monitor health trends associated with Alzheimer's disease and to determine the factors that prompt further research. The data collection involved participants who were right-handed and nearly evenly distributed by gender, with each individual having 3–5 distinct T1-weighted MRI scans collected during single sessions. The pre-processed MRI scans obtained were collated and cleaned with Microsoft Office Excel for further analysis.

3.2 Exploratory Data Analysis (EDA)

To thoroughly understand the longitudinal OASIS dataset and identify hidden relationships, trends, and statistical anomalies that could influence model implementation, an

Exploratory Data Analysis (EDA) was performed. Distributions and potential outliers were illustrated using descriptive statistics and visual methods, including scatter plots, box plots, and histograms.

Descriptive statistics (minimum and maximum values, mean, mode, quantiles, median, and variance) were calculated to summarize the dataset and help identify potential outliers, ensuring that these values fall within reasonable ranges. For example, MMSE scores in the longitudinal OASIS dataset were cross-verified with standard MMSE scores. To enhance the visual representation of the qualitative factors in the dataset, the EDA employed univariate analysis on the variables. By organizing the data using tables, graphs, and summary statistics, the information was communicated clearly, and potential skewnesses were identified.

Bivariate analysis was performed to evaluate the connections between the predictors and the cognitive scores. For example, a bar chart was created to display the distribution of MMSE scores across different age groups. This allows for a quick examination of the degree and pattern of their relationship, thus facilitating conclusions about the age distribution of the population alongside their cognitive scores.

3.3 Statistical Modeling and Strategy

This study will primarily focus on the quantitative method of multivariate data analysis, including statistical techniques and machine learning models, to develop predictive models that utilize brain structural changes to forecast Alzheimer's disease progression and assess the impact of demographic features like age, education, gender, and socio-economic status on brain structure and cognitive decline. Furthermore, the discussion of results incorporates a qualitative interpretive aspect, describing the clinical significance of the identified brain regions and model outcomes within the context of existing research and the practical implications for early diagnosis and patient treatment.

3.3.1 Traditional Models

Multivariate Analysis of Variance (MANOVA)

Multivariate Analysis of Variance (MANOVA) is a statistical technique that simultaneously compares group means across multiple dependent variables. By analyzing the variables together, MANOVA enables an assessment of differences across groups while accounting for correlations among the dependent variables. The null (H_0) and alternative (H_1) hypotheses are given as follows:

Null hypothesis (H_0): There is no significant difference in group mean vectors across the multiple dependent variables.

Alternative hypothesis (H_1): At least one group mean vector is significantly different from the others.

The outcome of MANOVA can be evaluated using commonly used test statistics, including Pillai's Trace (V), which measures the proportion of explained multivariate variation attributable to the group effect. It is calculated using the Treatment (Between) Sum of Squares and Cross-Products matrix ($\mathbf{B}$) and the Error (Within) Sum of Squares and Cross-Products matrix ($\mathbf{W}$), as follows:

$$V = \text{tr}\left(\mathbf{B}(\mathbf{B}+\mathbf{W})^{-1}\right) \tag{3.1}$$

$$\mathbf{B} = \sum_{e=1}^{g} n_e(\bar{x}_e - \bar{x})(\bar{x}_e - \bar{x})' \tag{3.2}$$

$$\mathbf{W} = \sum_{e=1}^{g}\sum_{i=1}^{n_e}(x_{ei} - \bar{x}_e)(x_{ei} - \bar{x}_e)' \tag{3.3}$$

where $\mathbf{B}$ represents the Treatment (Between) Sum of Squares and Cross-Products matrix, while $\mathbf{W}$ represents the Error (Within) Sum of Squares and Cross-Products matrix.

Pillai's Trace (V) measures the extent to which differences between groups explain variation in the combined dependent variables. Its value ranges from 0 to the smaller of the number of dependent variables and the numerator degrees of freedom. Larger values generally indicate greater differences between groups. However, the statistical significance of these differences is determined using the associated p-value. A p-value below the chosen significance level provides evidence against the null hypothesis, suggesting that the group mean vectors differ significantly. Other commonly used MANOVA test statistics include the Hotelling-Lawley Trace, Wilks' Lambda, and Roy's Largest Root.

Dependent variables, also known as response variables, are the variables whose values are examined to determine whether they differ in response to changes in the explanatory variables.

Independent variables, also known as explanatory variables or predictors, are variables used to explain or predict variations in the dependent variables.

A repeated measures MANOVA will be performed to assess whether the cognitive assessment scores jointly differ across independent groups.

3.3.2 Machine Learning Algorithms

Machine learning algorithms refer to models that allow computers to learn from data and make predictions or decisions without being explicitly programmed for every task. Both supervised and unsupervised learning models will be employed to analyze clinical demographic variables, cognitive assessment scores, and MRI-derived features to classify subjects into diagnostic categories and predict cognitive deterioration.

3.3.2.1 Supervised Machine Learning Algorithms

Supervised learning is a machine learning method that uses datasets with human-labelled inputs and outputs to train artificial intelligence models. The trained model identifies relationships between inputs and outputs, enabling it to predict outcomes accurately when presented with new, unseen data. The selected supervised learning models to be utilized in the study are Gradient Boosting, Support Vector Machine, Random Forest, Neural Network, and Naïve Bayes.

Gradient Boosting

This is an ensemble learning technique that produces accurate predictions by combining multiple decision trees into a single, more accurate model. The method builds models sequentially, with each successive model attempting to correct the errors made by previous models. Gradient boosting was developed within the framework of function approximation and numerical optimization by Friedman (2001). It is effective for predictive modelling because it can capture complex patterns and relationships within data.

The regularized objective function used in tree-based boosting methods such as XGBoost accounts for the empirical loss and model complexity (Chen and Guestrin, 2016):

$$\mathcal{L}(\theta) = \sum_{i=1}^{n} L(y_i, F(x_i)) + \sum_{k=1}^{K} \Omega(f_k), \qquad \Omega(f_k) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \tag{3.4}$$

where:

- $L(y_i, F(x_i))$ is the loss function measuring the difference between the observed and predicted values.
- $\Omega(f_k)$ is the regularization term that penalizes model complexity.
- T is the number of leaves in the tree.
- w_j is the weight of the jth leaf.
- γ and λ are regularization parameters.

In this research, gradient boosting will be considered because of its ability to model

complex relationships in datasets containing MRI-derived features, cognitive evaluation scores, and demographic clinical variables.

Support Vector Machine (SVM)

Support Vector Machine is a supervised machine learning algorithm used for classification and regression tasks. It identifies an optimal decision boundary, known as a hyperplane, that separates data points into their respective classes. SVM algorithms handle both linear and non-linear classification problems by using kernel functions to transform data into higher-dimensional spaces where separation may be possible (Cortes and Vapnik, 1995).

The model employs a constrained optimization problem to maximize the margin between the hyperplane and the support vectors:

$$\min_{\mathbf{w},b,\boldsymbol{\xi}} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}\xi_i \tag{3.5}$$

subject to:

$$y_i(\mathbf{w}^T\mathbf{x}_i + b) \geq 1 - \xi_i, \qquad \xi_i \geq 0, \quad \forall i.$$

where:

- $\mathbf{w}$ is the weight vector that determines the hyperplane.
- C is the regularization parameter that balances margin maximization and classification errors.
- ξ_i is the non-negative slack variable that permits misclassification.
- b is the bias term.

The SVM algorithm will be employed in this study to classify subjects as demented or non-demented based on MRI-derived features, cognitive evaluation scores, and demographic clinical variables. This will utilize both linear and non-linear SVMs to identify the optimal hyperplane that separates the different classes of data points.

Random Forest

Random Forest is an ensemble learning technique that combines the predictions of multiple decision trees to develop predictive models. It uses bootstrap aggregating (bagging) and random feature selection to improve model accuracy and stability. The decision trees are constructed using random samples of the dataset, with random subsets of features considered at each split. The final prediction is obtained by aggregating the predictions from the individual trees (Breiman, 2001).

The aggregate prediction function for B trees is given by:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{x}) \tag{3.6}$$

where T_b represents the prediction from the bth decision tree in the forest.

This study will apply Random Forest algorithms to develop a forecasting model for Alzheimer's disease progression using variables from the OASIS dataset, including MRI-derived features, cognitive assessment scores, and clinical demographic variables.

Naïve Bayes

Naïve Bayes is a supervised probabilistic machine learning algorithm derived from Bayes' theorem and used for classification problems. It determines the likelihood of a sample belonging to a particular class based on its observed features. The algorithm assumes conditional independence among the features given the class and selects the class with the highest posterior probability (Rish, 2001).

Bayes' theorem provides the mathematical foundation for updating the probability of a hypothesis when new evidence becomes available. The rule is given as:

$$P(C_k \mid X) = \frac{P(X \mid C_k)P(C_k)}{P(X)} \tag{3.7}$$

where:

- $P(C_k \mid X)$ is the posterior probability of a sample belonging to class C_k given its observed features.
- $P(X \mid C_k)$ is the likelihood of observing the features given class C_k.
- $P(C_k)$ is the prior probability of class C_k.
- $P(X)$ is the marginal probability of observing the features.

Under the conditional independence assumption, the predicted class is determined using the Maximum A Posteriori (MAP) rule:

$$\hat{C}(\mathbf{x}) = \arg\max_{C_k} \left[P(C_k) \prod_{j=1}^{n} P(x_j \mid C_k) \right] \tag{3.8}$$

In this study, the Naïve Bayes algorithm will be applied as a machine learning model to classify subjects into demented and non-demented groups using baseline MRI-derived features, cognitive assessment scores, and clinical demographic variables provided by the OASIS dataset.

Neural Network

A Neural Network is a machine learning model inspired by the interconnected structure of biological neurons. It consists of interconnected nodes, known as neurons, arranged in layers. The model learns patterns by adjusting weights and biases to transform input features into predictions or outputs. Neural networks generally comprise an input layer, one or more hidden layers, and an output layer. Information passes from one layer to the next in a feedforward neural network (Rumelhart et al., 1986).

The prediction function for a single hidden-layer neural network with K hidden units is given by:

$$\hat{y} = \sigma \left(\sum_{j=1}^{K} v_j g \left(\sum_{i=1}^{n} w_{ij} x_i + b_j \right) + c \right) \tag{3.9}$$

where:

- $\sigma(\cdot)$ is the activation function for the output layer.
- $g(\cdot)$ is the hidden-layer activation function.
- v_j is the weight linking the jth hidden unit to the output.
- w_{ij} is the weight connecting the ith input feature to the jth hidden unit.
- b_j is the bias term for the jth hidden unit.
- c is the bias term for the output unit.

This study will utilize a simple feedforward neural network to forecast cognitive decline using a layered network architecture based on the OASIS longitudinal dataset.

3.3.2.2 Unsupervised Machine Learning Algorithms

Unsupervised learning, in contrast to supervised learning, is a machine learning method that trains on unlabeled data to identify hidden patterns, similarities, and differences without human supervision. Unsupervised learning algorithms operate on input data to discover inherent patterns without prior knowledge of the outcomes. This research, in conjunction with supervised learning algorithms, will employ unsupervised learning methods to uncover patterns directly from the OASIS dataset, providing additional insights into the progression of Alzheimer's disease. Principal Component Analysis (PCA) and clustering will be conducted using baseline clinical demographic variables, cognitive assessment scores, and MRI-derived features from the OASIS longitudinal dataset.

Principal Component Analysis (PCA)

PCA is an unsupervised learning technique that reduces the dimensionality of a mul-

tivariate dataset by transforming correlated variables into fewer, linearly uncorrelated variables called principal components, while minimizing information loss. PCA enables the extraction of informative features from large datasets while preserving important statistical information from the original data, thereby addressing issues of multicollinearity and overfitting in regression analysis.

By PCA, one seeks to reduce p original variables to k new variables, where k is a compromise between the minimum number of variables and the maximum amount of information retained. The principal components are linear combinations of the original variables in the initial dataset.

We estimate the principal components such that the selected eigenvector maximizes the variance of the estimated principal component.

The optimization model for PCA is:

$$Y_p = \ell_{p1}X_1 + \ell_{p2}X_2 + \cdots + \ell_{pk}X_k$$

such that $\ell_{p1}, \ell_{p2}, \ldots, \ell_{pk}$ maximize $\mathrm{Var}(Y_p)$, subject to the constraint:

$$\|\boldsymbol{\ell}_p\|^2 = \boldsymbol{\ell}_p^T \boldsymbol{\ell}_p = \sum_{j=1}^{k} \ell_{pj}^2 = 1.$$

The resulting properties are:

$$\mathrm{Var}(Y_p) = \boldsymbol{\ell}_p^T \,\mathrm{Var}(\mathbf{X})\boldsymbol{\ell}_p = \boldsymbol{\ell}_p^T \boldsymbol{\Sigma} \boldsymbol{\ell}_p = \lambda_p \tag{3.10}$$

$$\mathrm{Cov}(Y_p, Y_q) = \boldsymbol{\ell}_p^T \boldsymbol{\Sigma} \boldsymbol{\ell}_q = 0, \qquad p \neq q \tag{3.11}$$

where λ_p is the eigenvalue (explained variance) of the pth principal component, $\boldsymbol{\ell}_p$ is its corresponding eigenvector, and $\boldsymbol{\Sigma}$ is the population variance-covariance matrix of the original variables $\mathbf{X}$.

Principal Component Analysis will be performed on the OASIS dataset to reduce the dimensionality of the clinical demographic variables, cognitive assessment scores, and MRI-derived features. This will allow the principal components to capture the maximum amount of variability in the data.

Clustering

Clustering is an unsupervised learning technique that involves dividing a dataset into groups, called clusters, based on similarities. It does so without using predefined labels or categories to reveal inherent patterns within the data. The K-means clustering algorithm

partitions the data into K clusters by assigning data points to the nearest cluster centroid through an iterative process (MacQueen, 1967; Somarribas and Loteanu, 2020).

The K-means algorithm minimizes the within-cluster sum of squared distances, given by:

$$J = \sum_{k=1}^{K} \sum_{i=1}^{n_k} \|\mathbf{x}_{ki} - \boldsymbol{\mu}_k\|^2 \tag{3.12}$$

where $\boldsymbol{\mu}_k$ is the kth cluster mean, $\mathbf{x}_{ki}$ is the ith data point assigned to cluster k, and n_k is the number of data points in cluster k.

Including K-means clustering in this study helps identify subgroups of subjects with similar MRI features and cognitive patterns, reducing the complexity of the longitudinal OASIS dataset.

3.4 Model Evaluation Metrics

These tools are crucial for evaluating the effectiveness of predictive models and classification tasks. A variety of performance evaluation metrics will be employed to verify the predictive accuracy of the Alzheimer's disease forecasting model and the classification outcomes, ensuring a comprehensive evaluation. The following metrics will be applied.

1. **Accuracy**

   This measures the overall correctness of the binary classification model, indicating whether a subject is demented or non-demented, using the baseline MRI and demographic data. The formula is given as:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} = \frac{TP + TN}{TP + TN + FP + FN} \tag{3.14}$$

   where:

   - TP is the true positive, where a demented subject is correctly classified as demented.
   - TN is the true negative, where a non-demented subject is correctly classified as non-demented.
   - FP is the false positive, where a non-demented subject is incorrectly classified as demented.
   - FN is the false negative, where a demented subject is incorrectly classified as non-demented.

2. **Precision (Positive Predictive Value)**

   This measures the proportion of correct positive predictions among all positive predictions made by the classification model, indicating how well the model avoids false alarms:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3.15}$$

3. **Sensitivity**

   This measures the model's capacity to identify actual dementia cases by assessing how effectively it recognizes illness when it actually exists:

$$\text{Sensitivity} = \frac{TP}{TP + FN} \tag{3.16}$$

4. **F1-Score**

   A measure of the harmonic mean of positive predictive value and sensitivity, showing the model's capacity to identify true dementia cases while minimizing false alarms:

$$F1\text{-Score} = 2 \times \frac{\text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \tag{3.17}$$

5. **Area Under Curve (AUC) and Receiver Operating Characteristic (ROC) Curve**

   The Receiver Operating Characteristic (ROC) curve illustrates the trade-off between the True Positive Rate (TPR), also referred to as sensitivity, and the False Positive Rate (FPR), which is calculated as:

$$TPR = \frac{TP}{TP + FN} \tag{3.18}$$

$$FPR = \frac{FP}{FP + TN} \tag{3.19}$$

   The Area Under the Curve (AUC) represents the area under the ROC curve and indicates how well the classification model distinguishes between the classes. Its value ranges from 0 to 1, with the following interpretation:

   - AUC = 0.5: no discrimination beyond random guessing.
   - AUC = 1: perfect discrimination.
   - AUC greater than 0.5: better discrimination than random guessing.

6. **Confusion Matrix**

   A confusion matrix is a tabular summary of the model's prediction results, comparing predicted classifications with actual classifications to visualize the model's performance.

7. **Silhouette Coefficient**

   The silhouette coefficient is a clustering evaluation measure that combines cluster cohesion and separation to assess how well data points fit within their assigned clusters (Rousseeuw, 1987). Cluster cohesion refers to the average distance of a data point from other points in its own cluster, while cluster separation refers to the average distance from that data point to points in the nearest neighbouring cluster. The silhouette coefficient ranges from -1 to 1, where:

   - -1: a poor assignment of a data point to a cluster.
   - 0: a data point lies near the decision boundary between neighbouring clusters.
   - 1: a strong assignment of a data point to its own cluster.

   The silhouette score is calculated as:

$$\text{Silhouette Score} = \frac{b - a}{\max(a, b)} \tag{3.20}$$

   where:

   - a is the mean distance between a data point and all other data points in its own cluster.
   - b is the lowest mean distance between the data point and all data points in any other cluster.

3.5 Research Ethics

The study exclusively uses a publicly available dataset from the Open Access Series Imaging Studies (OASIS). Since no sensitive personal information is included in the dataset used for this study, the research relies on the quality and integrity of the publicly available OASIS data. The dataset is used solely for scholarly purposes, contributing to research in basic and clinical neuroscience.

CHAPTER FOUR
Data Analysis, Results, and Discussions

4.0 Overview

This chapter presents the results of the data analyses employed to develop the predictive and classification models for Alzheimer's disease (AD). The analysis integrates both statistical methods and computational techniques to enhance our understanding of the relationships between demographic, cognitive, and MRI-derived factors concerning the onset and progression of Alzheimer's disease. It evaluates the effectiveness of the selected machine learning models, examines their outcomes, and relates the findings to the study's objectives. Ultimately, the results are analyzed in connection with previous research findings, highlighting whether the outcomes are consistent with or challenge the current literature on the progression and prediction of Alzheimer's disease.

4.1 Descriptive Analysis

Descriptive Statistics of the dataset

Descriptive statistics were computed to understand the traits of the dataset and summarize the quantitative variables of the subjects. This included various measures of central tendency, measures of variability, and quantiles, providing insights into the diversity of subjects and the overall structure of the data.

Table 4.1: Descriptive Statistics of Quantitative Variables: A Comprehensive Analysis

Variable	Mean	Median	SD	Q1	Q3	Min	Max
Visit	1.88	2	0.92	1	2	1	5
MR Delay	595.1	552	635.49	0	873	0	2639
Age	77.01	77	7.64	71	82	60	98

Variable	Mean	Median	SD	Q1	Q3	Min	Max
EDUC	14.6	15	2.88	12	16	6	23
SES	2.46	2	1.13	2	3	1	5
MMSE	27.34	29	3.68	27	30	4	30
CDR	0.29	0	0.37	0	0.5	0	2
eTIV	1488.13	1470	176.14	1357	1597	1106	2004
nWBV	0.73	0.73	0.04	0.7	0.76	0.64	0.84
ASF	1.19	1.19	0.14	1.1	1.29	0.88	1.59

Table 4.1 presents the summary statistics of the dataset's key features. The dataset included a substantial number of older adults, as indicated by a mean age of approximately 77 years, with most participants falling between 71 (Q1) and 82 (Q3) years old. This relates to the typical age range associated with Alzheimer's disease, with a sizeable portion of the sample exhibiting, on average, mild cognitive decline (mean MMSE score of 27.34). This suggests that the dataset represents an older population, comprising both individuals with dementia and those without. This age range aligns with the population most affected by Alzheimer's disease and mild cognitive impairment.

Visit numbers range from 1 to 5, with a mean of 1.88, indicating that most subjects had between one and two follow-up sessions. This confirms the longitudinal nature of the dataset but also suggests that repeated measures are relatively limited for many participants. MR Delay, which measures the time in days since the baseline scan, has a wide range (0–2639 days) and a high standard deviation of 635.49. This shows considerable variability in follow-up intervals, a common feature of real-world longitudinal clinical data.

Education (EDUC) averages 14.6 years (SD = 2.88), indicating that the sample generally has relatively high educational attainment, a factor known to influence cognitive reserve and the progression of dementia. Socioeconomic status (SES) exhibits moderate variability (mean = 2.46, SD = 1.13), with values spanning 1 to 5. This indicates a diverse sample in terms of social background, supporting generalizability.

MMSE scores range from 4 to 30, with a mean of 27.34 (SD = 3.68). Since the maximum MMSE score is 30, these values suggest that most participants are cognitively normal or mildly impaired. However, the lower tail (minimum = 4) confirms the presence of more advanced impairment cases. The Clinical Dementia Rating (CDR) has a mean of 0.29 (SD = 0.37), with scores from 0 to 2. The median of 0 indicates that most participants are nondemented at baseline, while the small number of higher scores (1–2) includes individuals with mild to moderate dementia.

In terms of MRI-derived variables, eTIV (estimated total intracranial volume) exhibits

significant variation (mean = 1488, SD = 176), reflecting the natural diversity in head sizes. The nWBV (normalized whole brain volume), with an average of 0.73 (SD = 0.04), suggests that participants preserve approximately 73% of their normalized brain volume, a finding that aligns with older adults, including those exhibiting early signs of neurodegeneration.

4.2 Exploratory Data Analysis Results

4.2.1 Distribution of Dementia Status

The dataset displays that non-demented cases are predominant among the subjects. At the same time, a lesser percentage experienced dementia, and an even smaller group underwent transition from a non-demented condition to dementia during the study's duration. This expected imbalance is indicative of the dataset's inclusion of a substantial number of normal aging adults, along with Alzheimer's disease-diagnosed subjects.

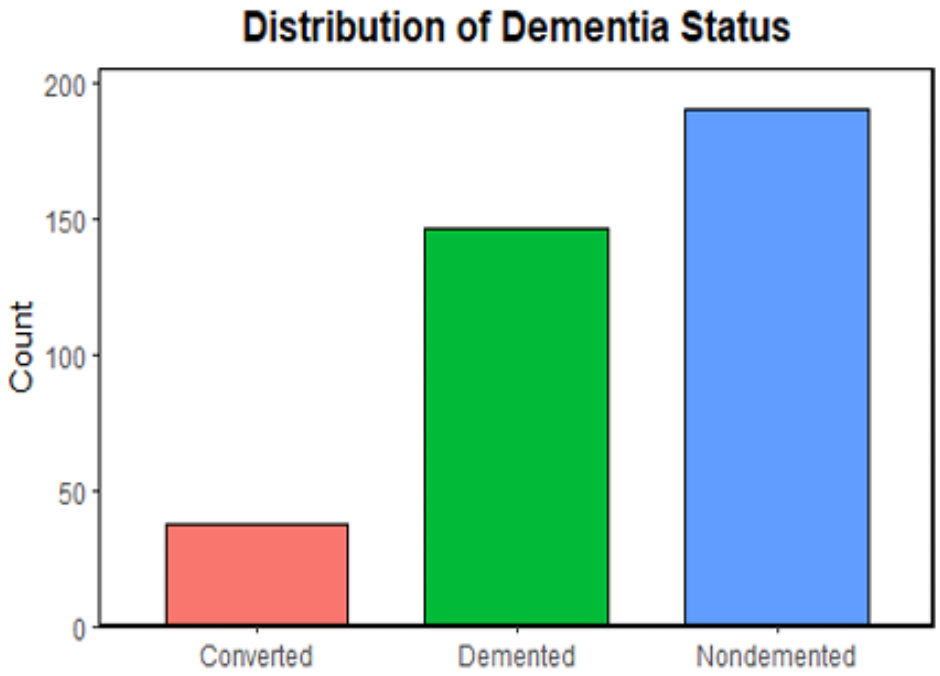


Figure 4.1: Bar chart of dementia status

4.2.2 Gender and Dementia Status Distribution: A Bivariate Analysis

A bivariate analysis of the dataset, considering the gender composition and dementia status of each subject, revealed a higher prevalence of dementia among males, despite the

larger proportion of females in the dataset. This pattern is inconsistent with the literature, as women are more likely to develop dementia, chiefly driven by their higher survival rate of old age, biological differences, and gendered social roles and responsibilities (Beam et al., 2018; Menzel et al., 2019; Andrew and Tierney, 2018).

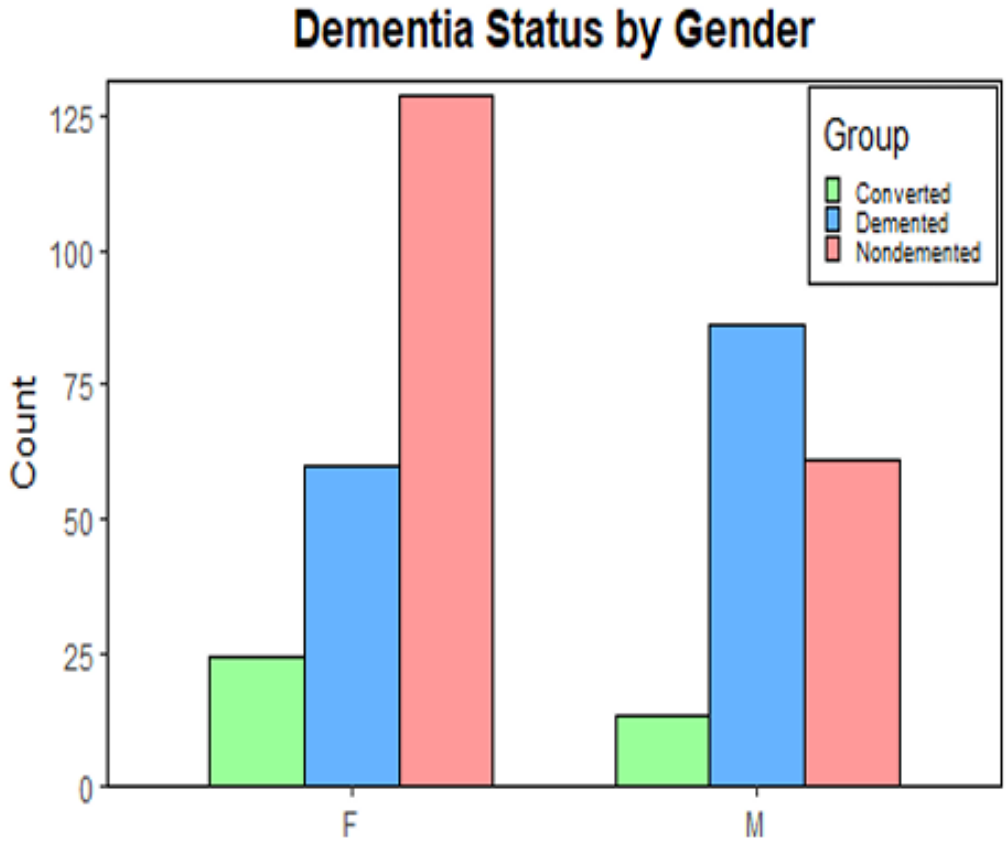


Figure 4.2: Bar chart illustrating gender differences in dementia status

4.2.3 A Bivariate Analysis of Age Disparities and Dementia Status

The age group analysis of the dementia status discovered that demented subjects were, on average, connected to advancing age, with age groups 71–80 and 81–90 displaying a higher frequency of demented cases. A similar observation is made about the converted subjects: the conversion rate increases with age, peaking in the 81–90 age group, followed by the 71–80 age group. This trend fortifies the well-established link in the literature and real-world situations between increasing age and the risk of developing Alzheimer's disease. Quite interestingly, fewer subjects were classified as demented and converted in the age group 91–100, indicative of other potential demographics in the dataset that might affect the dementia status.

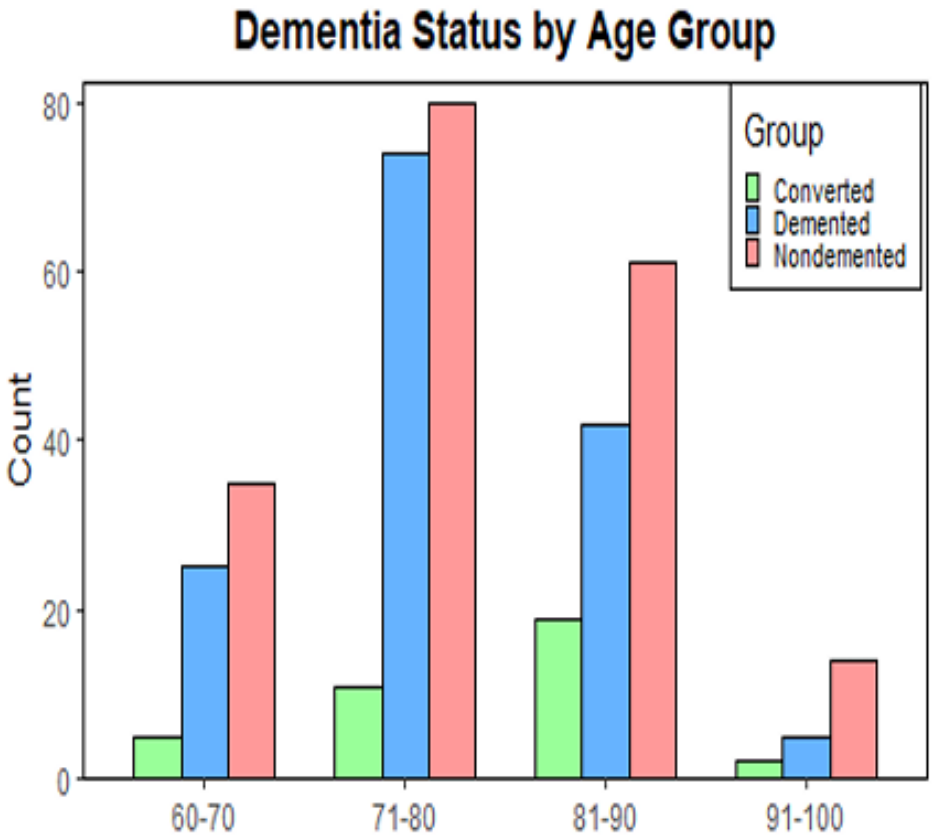


Figure 4.3: Bar chart illustrating dementia status across age groups

4.2.4 Bivariate Analysis of Cognitive Assessment Tools and Dementia Status

A box plot analysis of the cognitive evaluation tools (Mini-Mental State Examination scores and Clinical Dementia Rating scores) and the dementia classification depicts a coherent and uniform association.

Figure 4.4 shows the Mini-Mental State Examination (MMSE) scores over the dementia status categories. The analysis reveals that the MMSE scores were higher among non-demented subjects, demonstrating normal cognition, followed by subjects who transitioned from a healthy state to a demented state. Subjects classified as demented, however, showed considerably lower MMSE scores, signaling a notable decline in cognitive abilities. This trend is expected and consistent with public health research and global trends.

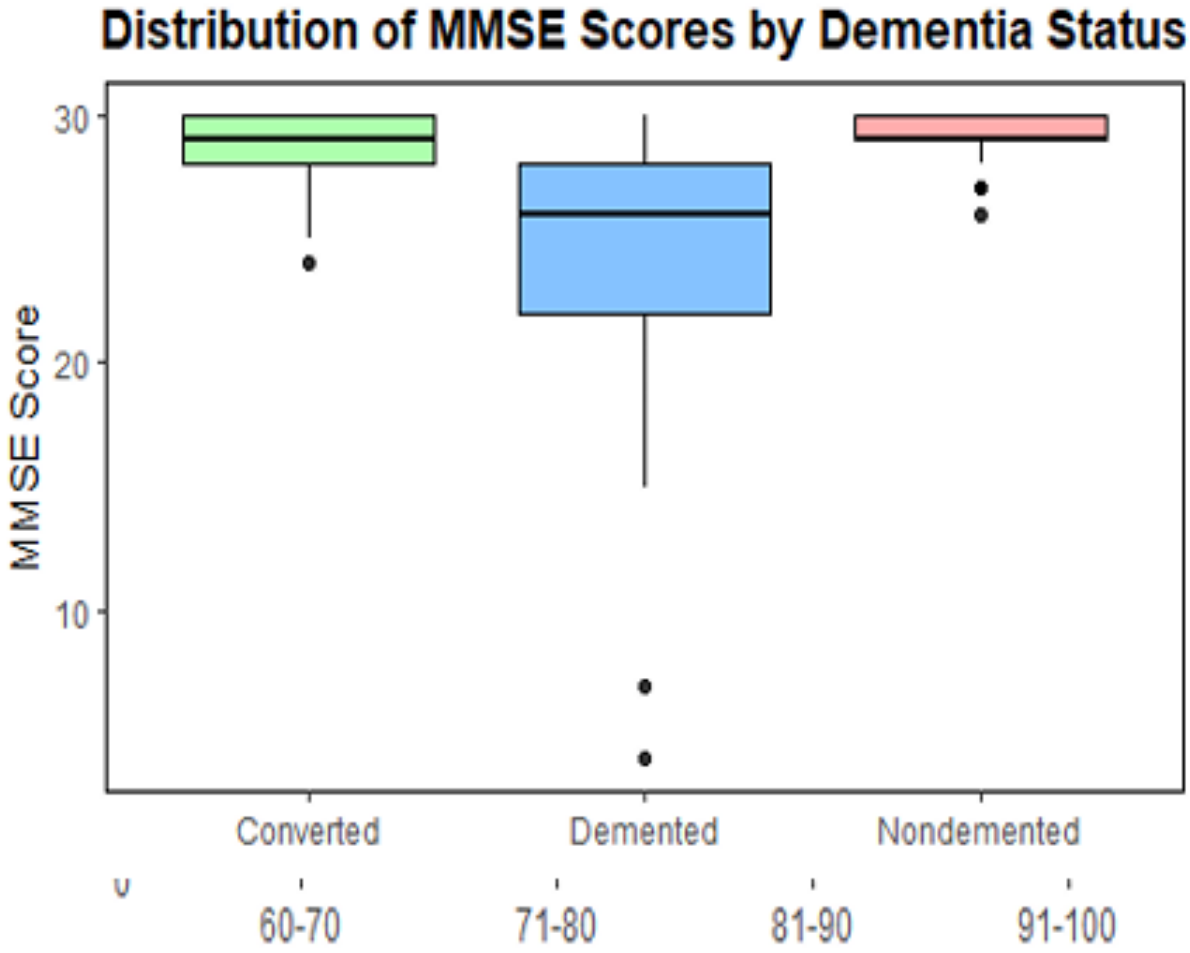


Figure 4.4: Box plot illustrating the distribution of MMSE scores by dementia status

The Clinical Dementia Rating (CDR) box plot analysis in Figure 4.5 displays lower CDR scores for non-demented participants, with an approximate CDR score of zero. Converted subjects exhibited moderate CDR scores, ranging from 0 to 0.5, indicating a gradual

progression in cognitive decline over the study period. CDR scores were relatively high in dementia-classified subjects, indicating a decline in neurodegenerative conditions and deterioration in functionality.

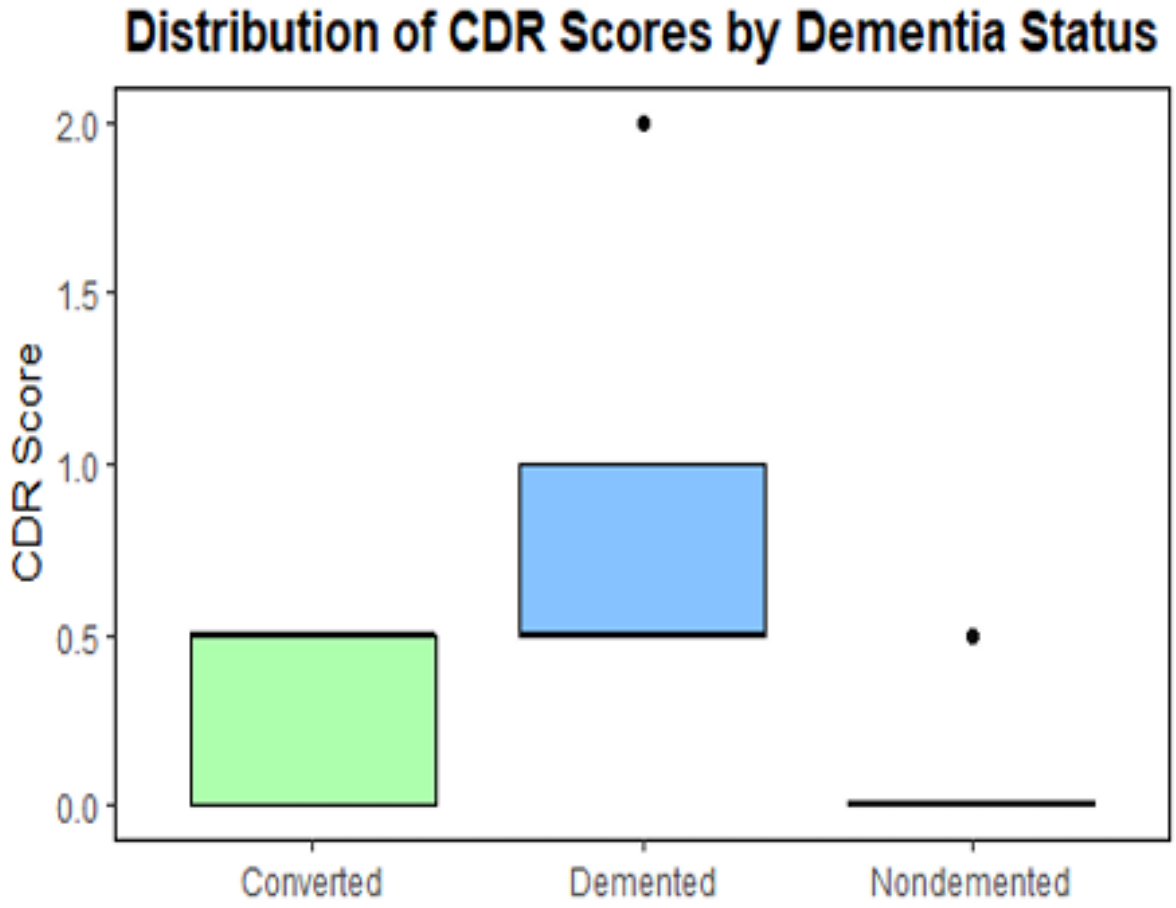


Figure 4.5: Box plot illustrating the distribution of CDR scores by dementia status

4.2.5 Dementia Status by Socioeconomic Disposition: A Bivariate Analysis

The socioeconomic disposition of subjects clearly and somewhat oppositely influenced dementia status in the dataset. Participants with elevated socioeconomic scores showed significantly greater levels of dementia; those rated four in economic and social standing were most affected, followed by those rated three. Non-demented individuals were more common in lower socioeconomic categories, with grade two subjects showing the highest proportion of non-demented cases. This pattern conflicts with existing research and global neuroscience data, which indicate that individuals with lower socioeconomic status (SES) are more likely to develop dementia. Subjects who shifted from healthy cognition to progressive neurodegeneration during the study demonstrated an inverse relationship

with high socioeconomic status ratings.

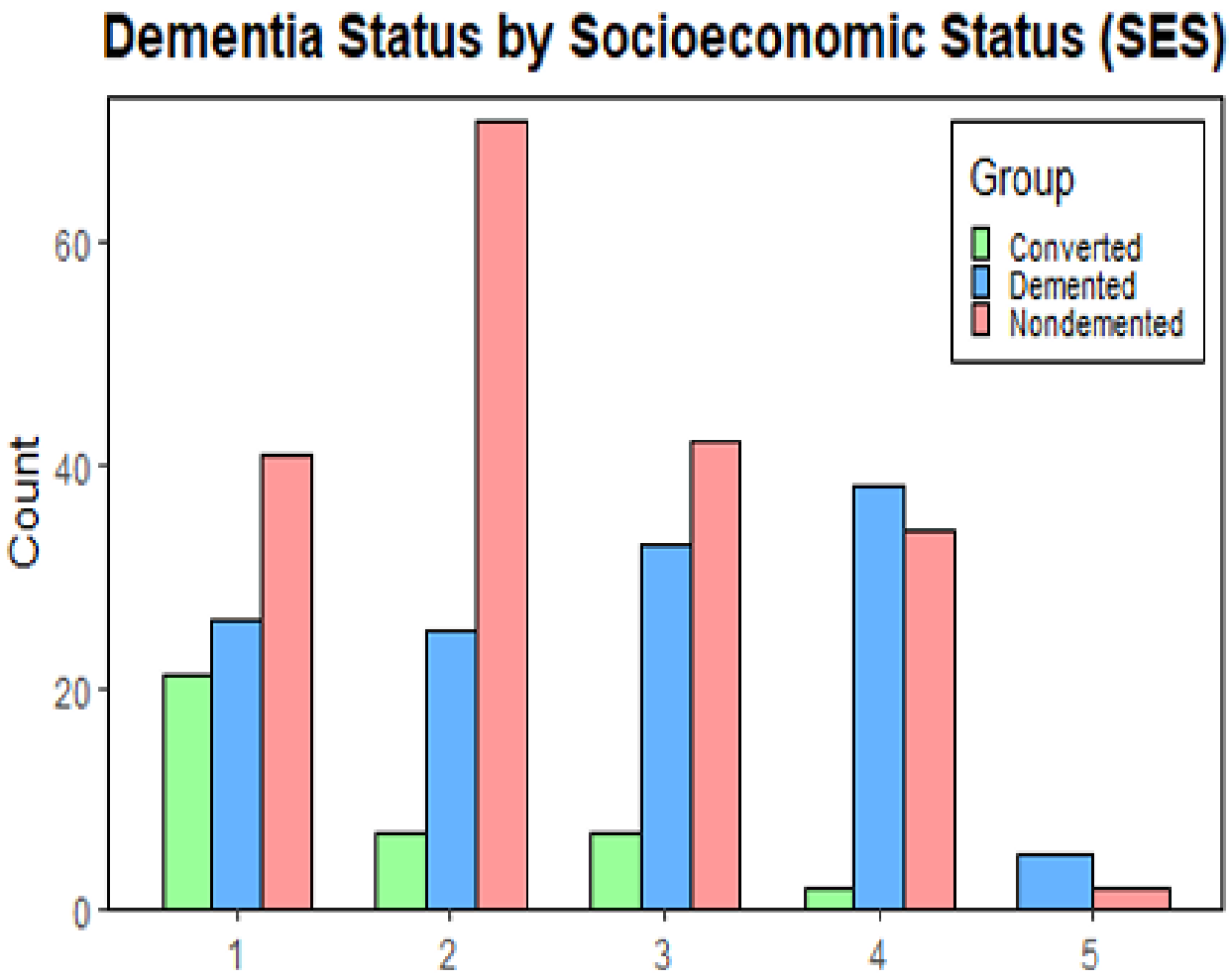


Figure 4.6: Bar chart illustrating dementia status across socioeconomic status categories

4.2.6 Analysis of Dementia Status Based on Education: A Bivariate Examination

The influence of education level on dementia classification in the dataset is highly significant. A bivariate bar chart analysis of dementia categories and educational levels revealed that higher education levels are associated with a lower prevalence of dementia among the subjects. Participants with the highest level of education (21 to 23 years of schooling) had the fewest cases of dementia, while non-demented individuals were more common at higher education levels. This trend aligns with established research and worldwide observations, as higher educational attainment generally provides a greater safeguard against cognitive impairments, a concept clearly articulated by the Cognitive Reserve Theory (Stern, 2007).

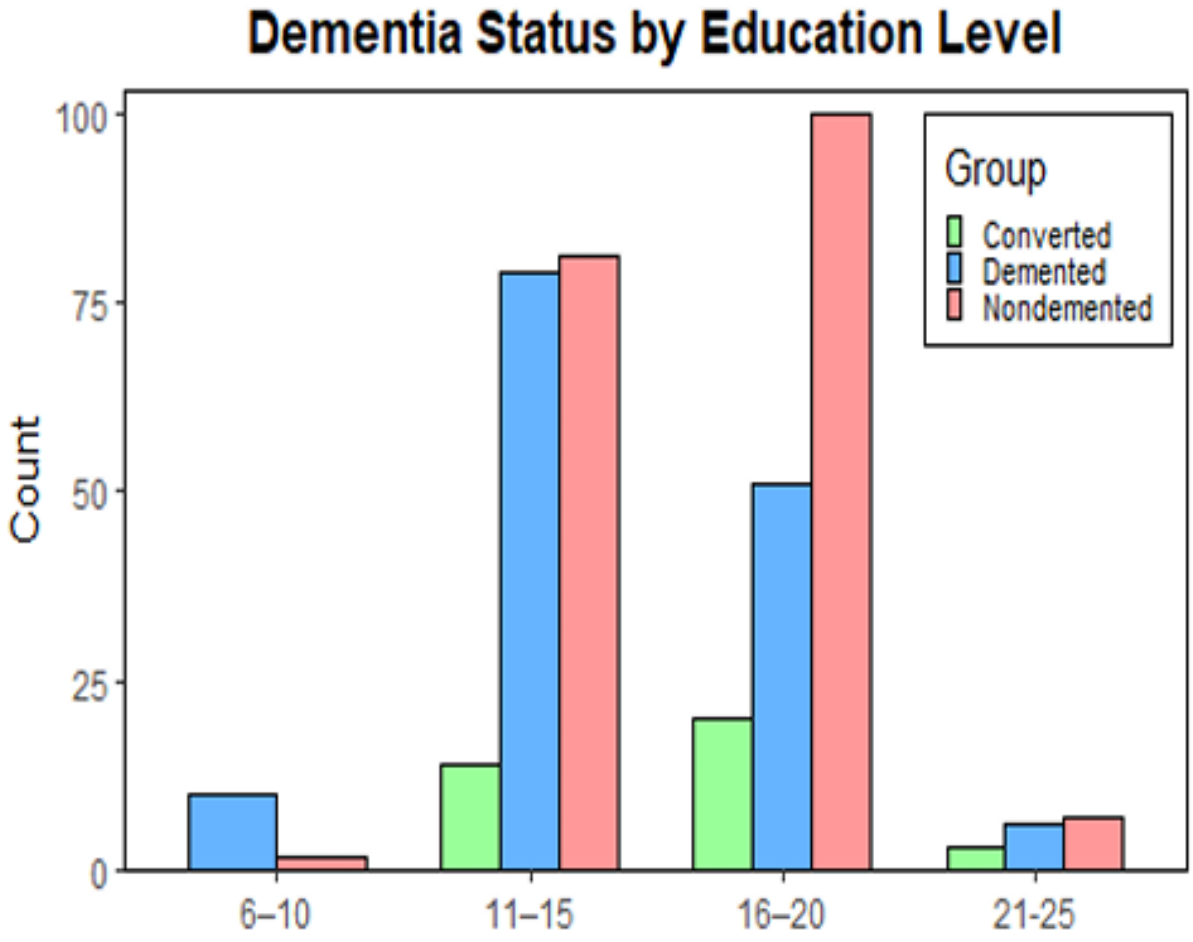


Figure 4.7: Bar chart illustrating dementia status across education levels

4.2.7 MRI-derived Features and Cognitive Status

An analysis using a bivariate box plot of the normalized Whole Brain Volumes (nWBV), categorized by dementia status, revealed a consistent and uniform relationship between the two. Normalized whole-brain volumes were relatively higher among non-demented subjects, indicating a greater amount of intact brain tissue. In comparison, participants classified as demented exhibited noticeably lower normalized whole-brain volumes. This reflects the measurable shrinkage of notable brain regions. Subjects who metamorphosed from a healthy condition to one of dementia exhibited moderate nWBV values, indicating a progressive reduction in brain volume. This trend is expected and aligns with existing literature; the Atrophy-Cognition Model, which posits progressive neurodegeneration in distinct brain regions of an AD patient, corresponds to a gradual decline in cognition.

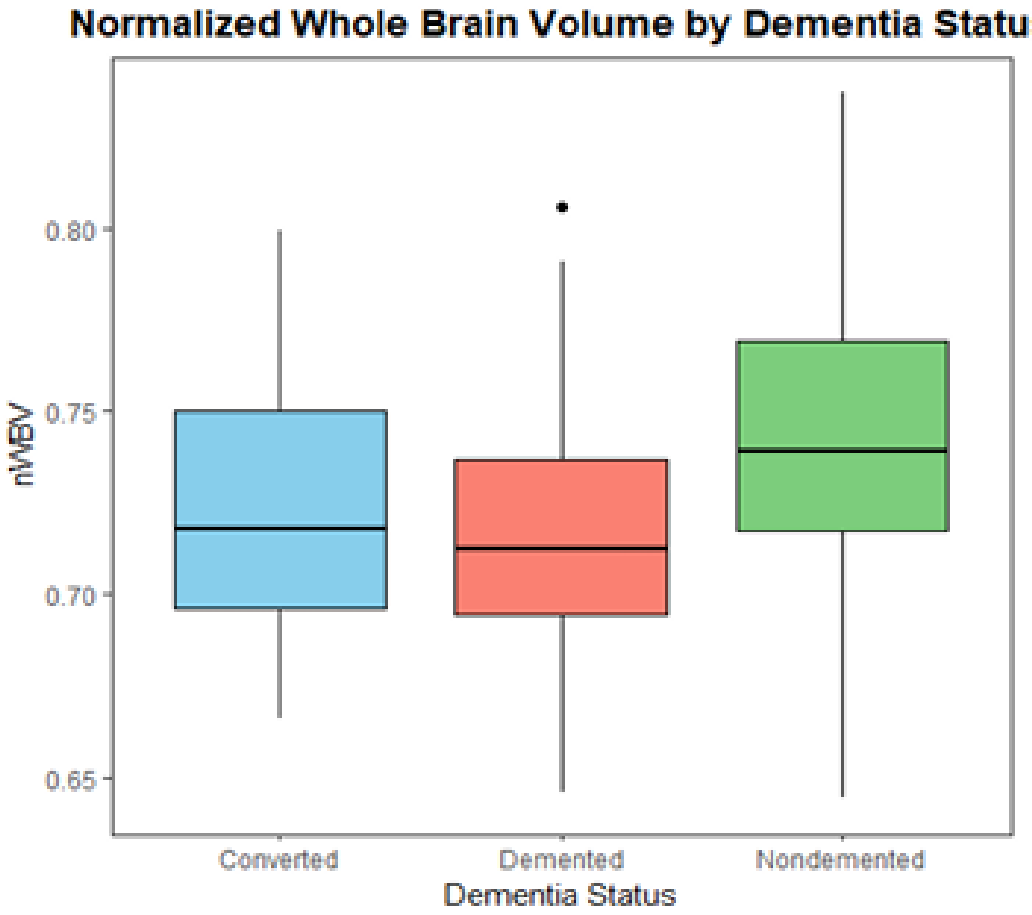


Figure 4.8: Box plot illustrating normalized whole-brain volume by dementia status

4.2.8 Correlation Analysis and Multivariate Relationships

A Pearson correlation analysis, used to identify multivariate relationships and spot patterns within the dataset, uncovered the following key insights:

- The Mini-Mental State Examination (MMSE) showed a significant negative correlation with the severity of dementia, as measured by the Clinical Dementia Rating (CDR):

$$r \approx -0.71.$$

  This finding aligns with clinical expectations, as more advanced stages of dementia generally lead to lower scores on the MMSE.

- Normalized whole-brain volume revealed a mild negative correlation with dementia severity (CDR):

$$r \approx -0.35.$$

  This suggests that individuals with smaller nWBV values are more likely to have high dementia severity.

- Age and normalized whole-brain volume (nWBV) exhibited an intermediate negative correlation:

$$r \approx -0.53.$$

  Older adults generally have smaller normalized whole-brain volumes.

- Atlas Scaling Factor (ASF) correlates almost negatively perfectly with the Estimated Total Intracranial Volume (eTIV):

$$r \approx -0.99.$$

  This demonstrates a mathematical relationship, where the eTIV is calculated by dividing a constant by the ASF.

- Dementia severity, as assessed by CDR, tends to have a very weak individual correlation with variables such as socioeconomic status (SES), atlas scaling factor (ASF), age, education (EDUC), and estimated Total Intracranial Volume (eTIV). They, however, establish a robust statistical basis for the predictive modelling and MANOVA analysis, guaranteeing that all relationships are valid when combined with other predictors.

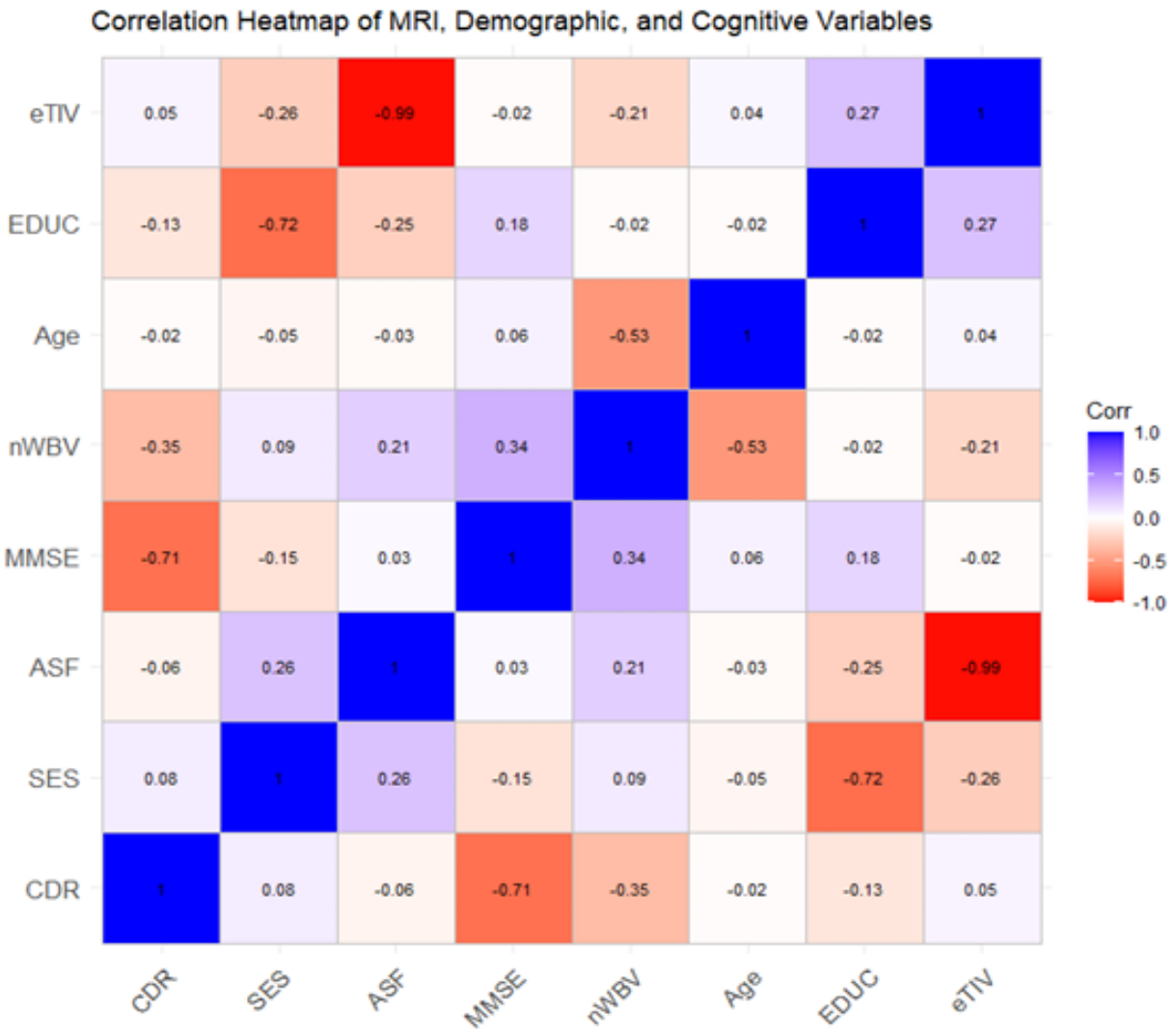


Figure 4.9: Correlation heatmap illustrating the relationship between variables

4.3 Multivariate Analysis of Variance (MANOVA) Results

A multivariate analysis of variance (MANOVA) was conducted to investigate whether there was a significant difference in the cognitive scores and MRI-derived features across the dementia status, as measured by the Group. A preliminary assumption testing conducted revealed the following key insights:

- **Multicollinearity:** The correlation coefficients were less than 0.9; thus, the assumption of multicollinearity is tenable.
- **Independence:** A one-way repeated measures MANOVA was conducted to suit the non-independent observations, given the longitudinal structure of the dataset.
- **Normality:** The Shapiro-Wilk test for the dependent variables showed a breach of the normality requirement ($p < 0.01$). Nevertheless, due to the large sample size in each group, the robustness of MANOVA is adequate to account for this violation, in line with the Central Limit Theorem.
- **Outliers:** While boxplot analysis on the variables indicated potential univariate outliers, all cases were retained since all the values fall within the clinically standard range of the variables. The presence of a few multivariate outliers was also detected using the Mahalanobis distance (a total of 2 cases at a one-percent significance level). Considering the large sample size and the diversity in clinical characteristics, all cases were included in the analysis.
- **Linearity:** The assumption of linearity is satisfactory per the inspection of the scatterplot matrix.
- **Variance-covariance homogeneity:** The Box's M test did not meet the requirement of homogeneity of the variance-covariance matrix ($p < 0.01$). Consequently, Pillai's Trace is the preferred test statistic to employ, as it has demonstrated greater resilience to these breached assumptions.

Table 4.2: Repeated Measures MANOVA Results Using Pillai's Trace

Effect	**Pillai's Trace**	**df (num., den.)**	**F**	**p-value**
Group	0.446	6, 694	33.240	< 0.001
Visit	0.054	3, 346	6.533	0.00026
Group $\times$ Visit	0.081	6, 694	4.883	< 0.001

The outcome of the RM-MANOVA revealed a statistically significant multivariate difference among the groups (demented, non-demented, and converted) on the combined dependent variables (MMSE, eTIV, and nWBV), Pillai = 0.446, $F(6, 694) = 33.240$,

$p < 0.001$. The null hypothesis was therefore rejected, leading to the conclusion that dementia status correlates with observable variations in brain structure and cognitive abilities. A follow-up univariate ANOVA was conducted at a five-percent alpha level.

The result indicated statistically sufficient evidence to reject the null hypothesis that the MMSE scores and nWBV values do not differ across the dementia status, $F(2, 356) = 130.658$, $p < 0.001$, partial $\eta^2 = 0.42$, and $F(2, 356) = 21.381$, $p < 0.001$, partial $\eta^2 = 0.11$, respectively, concluding that dementia status is related to MMSE scores and nWBV values. The effect size was large in both variables. Outcome of the univariate ANOVA on eTIV revealed that eTIV values did not differ across dementia status with a small effect size, $F(2, 356) = 0.657$, $p = 0.519$, partial $\eta^2 = 0.004$.

4.4 Principal Component Analysis (PCA) Results

A principal component analysis was conducted on the numeric variables (Age, MMSE, nWBV, eTIV, Visit, MR Delay, EDUC, and SES) to decrease the dimensionality by uncovering hidden patterns in the dataset and converting correlated variables into a smaller set of linearly uncorrelated variables, all while preserving data integrity. Before conducting the analysis, the dataset was normalized to guarantee that all variables were on a consistent scale.

The analysis's results showed that the initial five principal components account for a significant portion of the variation (91.7%), leading to their retention for further examination and interpretation. Component 1 explains (27.68%) of the variation, component 2 explains (23.13%), component 3 explains (18.5%), component 4 explains (12.79%), and component 5 accounts for (9.61%) of the variation.

The final three principal components account for negligible information (8.3%), and hence their rejection.

Analysis of the component loadings revealed that the initial component primarily captures visit number and the time (delay) between different MRI scans in a positive light. The second component positively correlates with socioeconomic disposition, while negatively correlating with educational gradient. PC3 primarily signifies the negative impact of brain volume integrity and cognitive score, whereas PC4 appears to be negatively associated with cognitive function. PC5 shows a negative correlation with intracranial volume. The variable loadings for the initial four principal components are shown in the table below.

A 3D representation of the PCA outcomes based on dementia status provided a visual overview of both demented and non-demented individuals across the first three principal components. The 3D PCA visualization (as shown in Fig. 4.11) revealed a mild distinction between participants with dementia and healthy subjects, with significant overlap

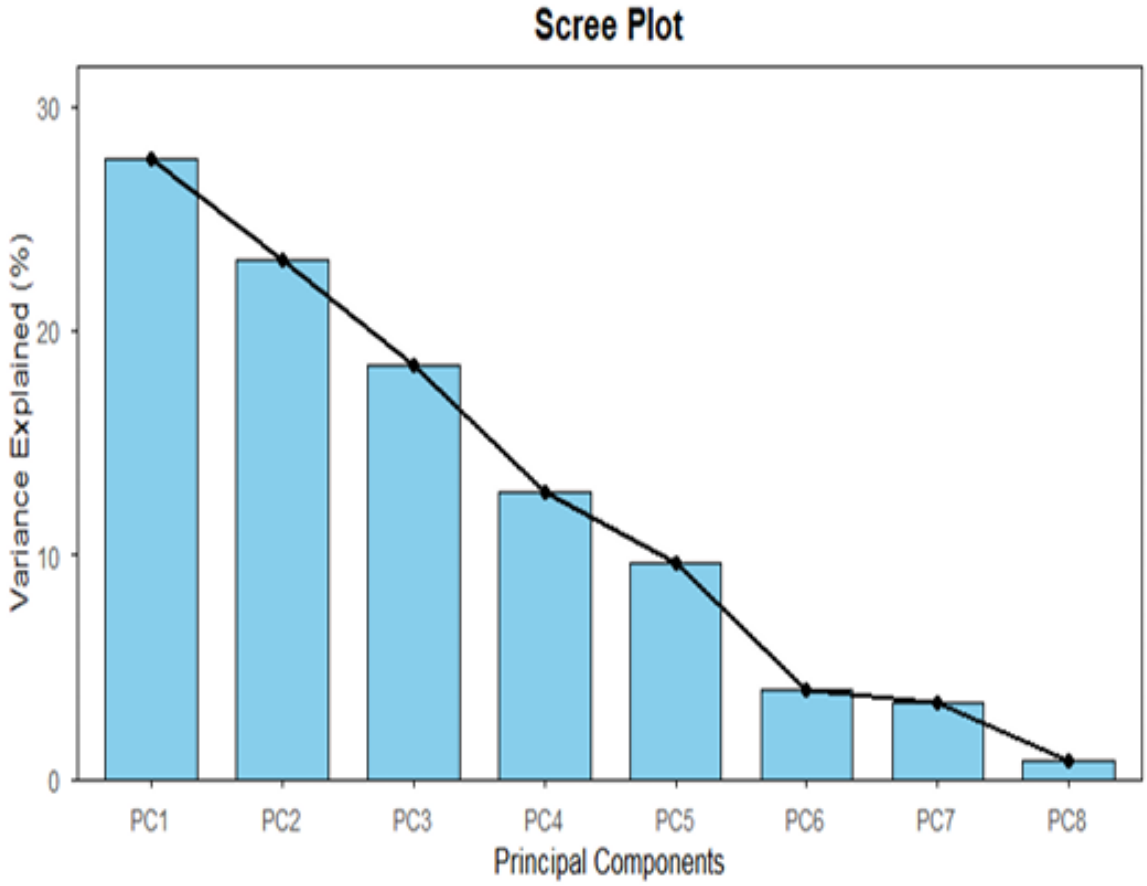


Figure 4.10: A scree plot illustrating the principal components along with the percentage of variance accounted for

Table 4.3: Component Loadings

Variables	**Components**				
	PC1	**PC2**	**PC3**	**PC4**	**PC5**
Visit	0.528	0.281	-0.339	0.148	-0.105
MR Delay	0.528	0.264	-0.377	0.064	-0.058
Age	0.32	0.185	0.368	-0.626	0.056
EDUC	0.267	-0.592	0.009	0.031	-0.264
SES	-0.296	0.565	-0.073	0.002	0.308
MMSE	0.02	-0.263	-0.422	-0.654	0.391
eTIV	0.293	-0.226	0.178	0.386	0.814
nWBV	-0.307	-0.164	-0.627	0.063	0.053

observed. For clarity purposes and easy interpretability, subjects who transitioned from a healthy state to a demented condition were excluded. This reinforces the argument that the retained components extract significant information from the dataset in a reduced dimension.

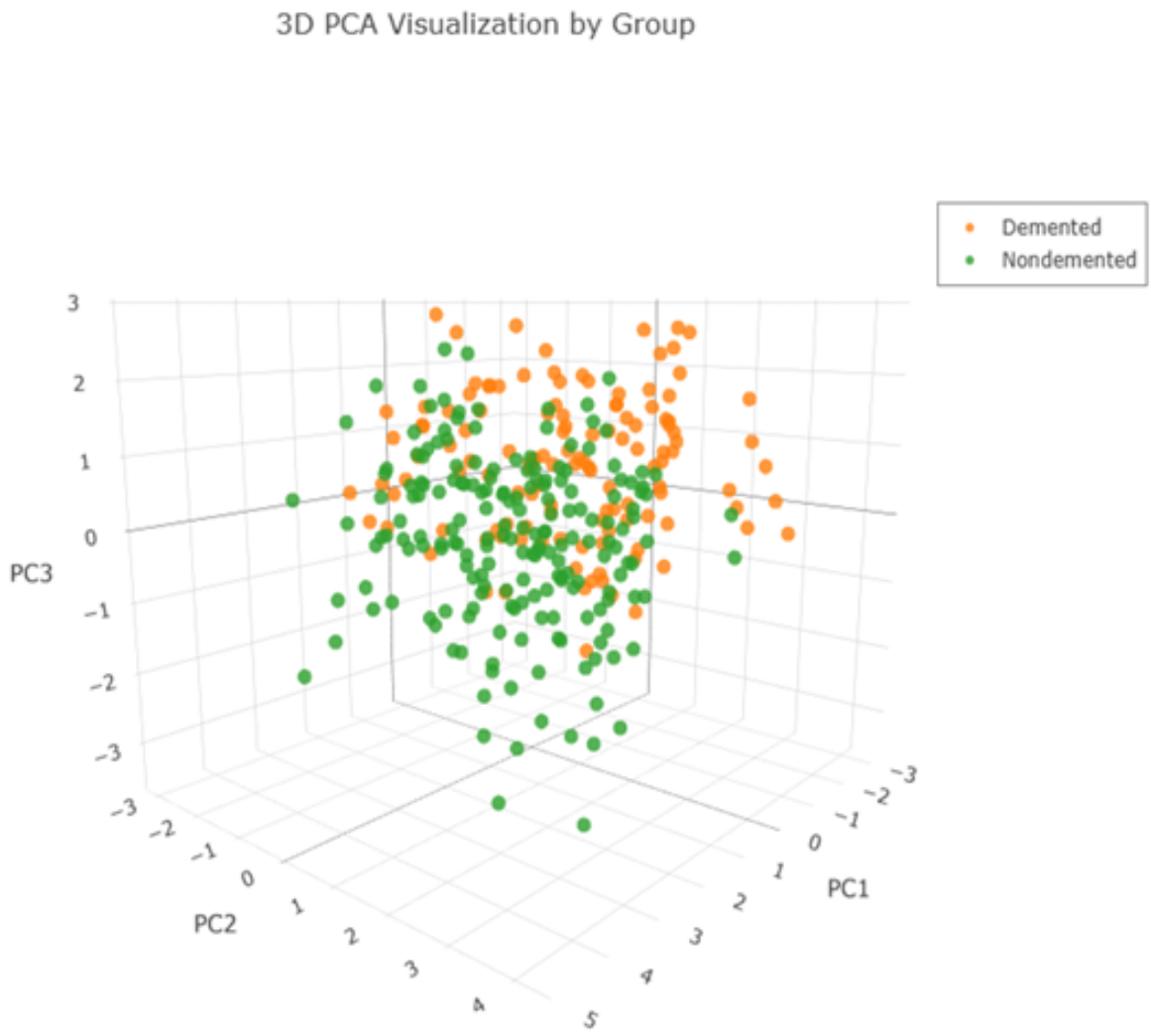


Figure 4.11: 3D Visualization of principal components by dementia status

4.5 K-Means Clustering Utilizing Principal Components

A K-means clustering analysis was performed to categorize the observations into similar groups, following the dimensionality reduction through Principal Component Analysis. This analysis indicated that the first five principal components effectively capture a significant amount of information in the dataset. The optimal number of clusters (k) was identified using both the Elbow method and the Silhouette score. The Elbow method,

based on the Within-Sum-of-Squares (WSS) plot, suggested an optimal number of four clusters ($k = 4$), indicating that the retained PCA-transformed data could be divided into four distinct subgroups. However, the Silhouette score analysis identified five clusters ($k = 5$) as the optimal solution, suggesting that this configuration provides a more suitable clustering structure. Consequently, five clusters were selected for the analysis based on the Silhouette score, which provided the stronger basis for the final clustering solution.

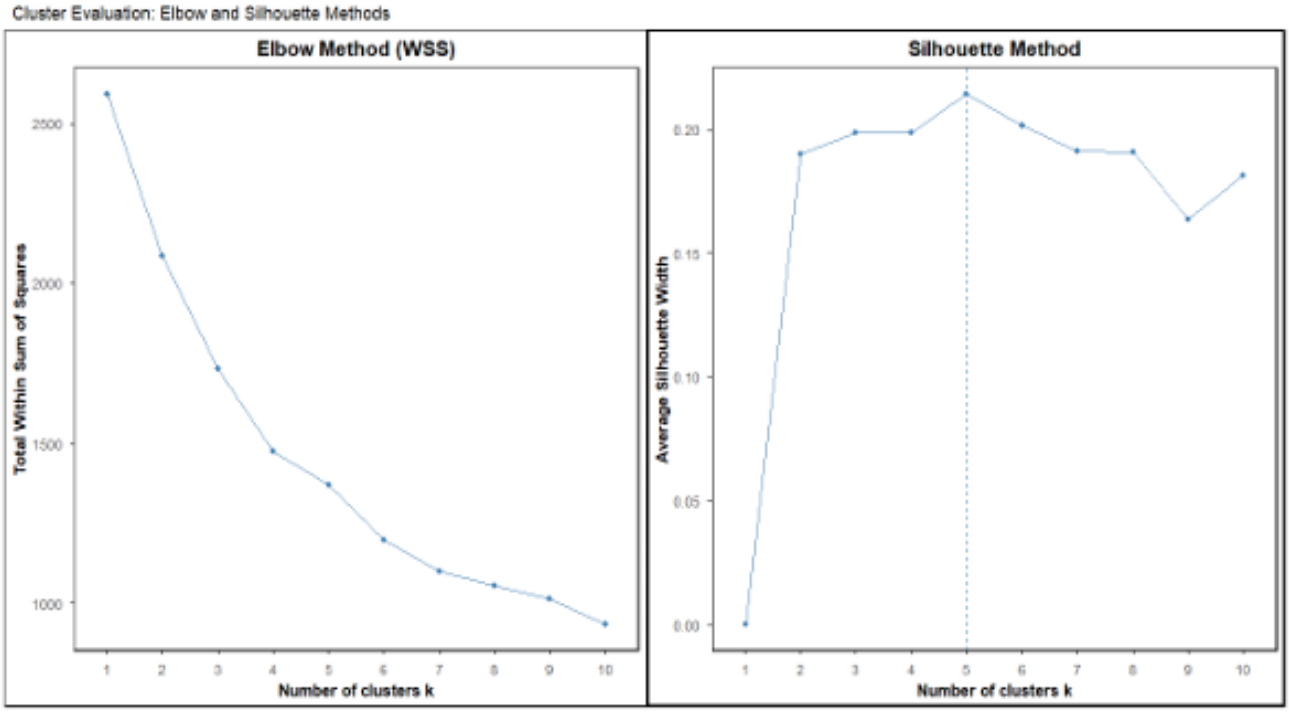


Figure 4.12: Comparative cluster evaluation

A two-dimensional graphic representation of the K-means clustering (Fig. 4.13), based on the first four principal components and $k = 5$, revealed five clearly defined groups based on similarities in brain structure, cognitive characteristics, and clinical demographics, with significant overlaps. The cluster effectiveness was assessed using the silhouette coefficient, yielding an average silhouette score of 0.214, indicating a weak but discernible clustering structure, which is consistent with the clinical dataset in the existing literature.

4.6 Developing Models to Forecast and Classify Dementia Progression

This section outlines the result of the predictive modeling phase, concentrating on the effectiveness of the five machine learning algorithms: Gradient Boosting, Support Vector Machine (SVM), Random Forest, Neural Network, and Naïve Bayes. We aim to assess

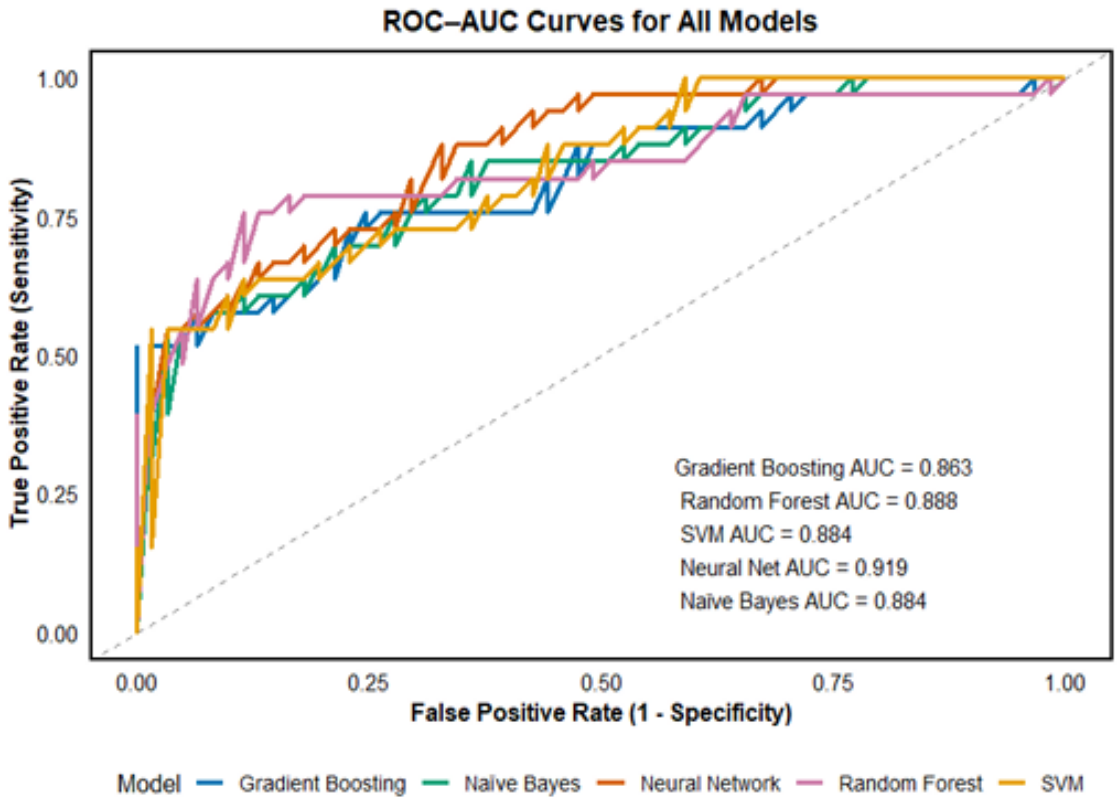


Figure 4.13: K-means clustering with five clusters based on the first five principal components

how well these models predict dementia progression, thereby determining which approach best captures the relationship between the response variable (Clinical Dementia Rating) and the predictors (retained principal components and gender). It is worth noting that the Synthetic Minority Over-sampling Technique (SMOTE) was employed to address the high-class imbalance in the dataset.

4.6.1 Comparison of Model Performance

Table 4.4 offers a comparative overview of the effectiveness of all models used in developing the predictive model, highlighting their test accuracies and cross-validation scores.

Table 4.4: Summary Comparison of the Performance Across All Predictive Models

Model	**Test Accuracy**	**CV Accuracy (±SD)**	**CV F1 Macro (±SD)**
Gradient Boosting	74%	75.3% (±4.2%)	74.9% (±13.5%)
Random Forest	76%	73.6% (±5.4%)	69% (±21.5%)
SVM	72.1%	75% (±5.6%)	73.9% (±15.4%)
Neural Network	75%	75% (±4.4%)	70.8% (±21.1%)
Naïve Bayes	74%	70.9% (±4.6%)	71.5% (±19.9%)

4.6.2 Gradient Boosting

The Gradient Boosting model demonstrated superior performance on the test dataset. Since the cross-validation results are roughly comparable to the test accuracy, this indicates the model's stability and suggests it performs pretty well on new data.

- **Test Accuracy:** 74%
- **Cross-Validation Accuracy:** 75.3% (±4.2%)
- **Cross-Validation F1 (Macro):** 74.9% (±13.5%)

However, careful tuning is necessary to attain optimal performance.

4.6.3 Random Forest

The Random Forest algorithm achieved the highest test accuracy, reflecting its effectiveness in practical applications. However, its cross-validation accuracy was lower than that of Gradient Boosting, while the significant variability in its cross-validation F1 score suggests that the model is inconsistent and sensitive to the training data.

- **Test Accuracy:** 76%
- **Cross-Validation Accuracy:** 73.6% (±5.4%)
- **Cross-Validation F1 (Macro):** 69% (±21.5%)

Despite considerable generalization, the model still struggled with cases of high-severity dementia, suggesting that class imbalance remained a problem, even after applying SMOTE resampling.

4.6.4 Support Vector Machine (SVM)

The SVM classifier demonstrated a moderate level of performance, albeit comparatively lower, on the test set. The model exhibits a good balance between precision and recall, as demonstrated by its consistent performance in cross-validation.

- **Test Accuracy:** 72.1%
- **Cross-Validation Accuracy:** 75% (±5.6%)
- **Cross-Validation F1 (Macro):** 73.9% (±15.4%)

Further fine-tuning the model's parameters could enhance its performance.

4.6.5 Neural Network

The Neural Network algorithm demonstrated a competitive level of accuracy in testing, with perfect consistency between the test set and the validation set. The Neural Network classifier achieved a comparatively lower cross-validation F1 score, which reflects its inconsistency in balancing false positives and false negatives within the severe dementia category.

- **Test Accuracy:** 75%
- **Cross-Validation Accuracy:** 75% (±4.4%)
- **Cross-Validation F1 (Macro):** 70.8% (±21.1%)

4.6.6 Naïve Bayes

The Naïve Bayes algorithm yielded a moderate improvement in test results, with an acceptable level of consistency in accuracy across both the test and validation sets.

- **Test Accuracy:** 74.0%
- **Cross-Validation Accuracy:** 70.9% (±4.6%)
- **Cross-Validation F1 (Macro):** 71.5% (±19.9%)

4.6.7 Comparative Analysis

- The Random Forest model achieved the highest overall accuracy on the test set at 76.0%, outperforming the Neural Network, which reached a test accuracy of 75.0%,

as well as Naïve Bayes and Gradient Boosting at 74.0%. SVM achieved the lowest accuracy at 72.1%.

- Although Random Forest demonstrated the highest test accuracy, all models continued to face difficulties with the severely demented category, emphasizing that class imbalance remained a significant issue despite the application of SMOTE resampling.
- In the process of cross-validation, Gradient Boosting achieved the highest accuracy at 75.3%, with SVM and Neural Network not far behind at 75.0%. Random Forest followed with an accuracy of 73.6%, and Naïve Bayes came in at 70.9%.

4.7 Model Comparison Metrics

Table 4.5 below presents a comparative summary of the performance across all models. The Gradient Boosting and Random Forest models exhibited outstanding performance on the merged dataset, whereas the SVM, Neural Network, and Naïve Bayes models attained a moderate level of overall performance.

Table 4.5: Model Comparison Metrics

Model	**Accuracy**	**Precision (Macro)**	**Recall (Macro)**	**F1-score (Macro)**
Gradient Boosting	0.926	0.913	0.919	0.911
Random Forest	0.931	0.908	0.923	0.912
SVM	0.774	0.752	0.703	0.714
Neural Network	0.777	0.773	0.738	0.729
Naïve Bayes	0.736	0.743	0.689	0.705

4.7.1 Receiver Operating Characteristic–Area Under the Curve (ROC–AUC) Analysis

The models' predictive performances were assessed using the Receiver Operating Characteristic–Area Under the Curve (ROC–AUC) metric, which provides a more comprehensive evaluation than just accuracy. The ROC–AUC curve illustrates the connection between each model's True Positive Rate (TPR) and False Positive Rate (FPR) across all decision thresholds. To encapsulate the results across the four classifications of dementia severity, a micro-average was computed. The diagonal dashed line symbolizes random guessing and serves as a baseline, with an AUC of 0.50.

The ROC-AUC assessment revealed an outstanding differentiation of dementia severity, indicated by AUC values exceeding 0.85 across all models. Neural Networks demonstrated the strongest discriminative power, with an AUC of 0.919, while Random Forest followed

with an AUC of 0.888. SVM and Naïve Bayes were not far behind, achieving AUCs of 0.884 each. Gradient Boosting produced a moderate AUC of 0.863.

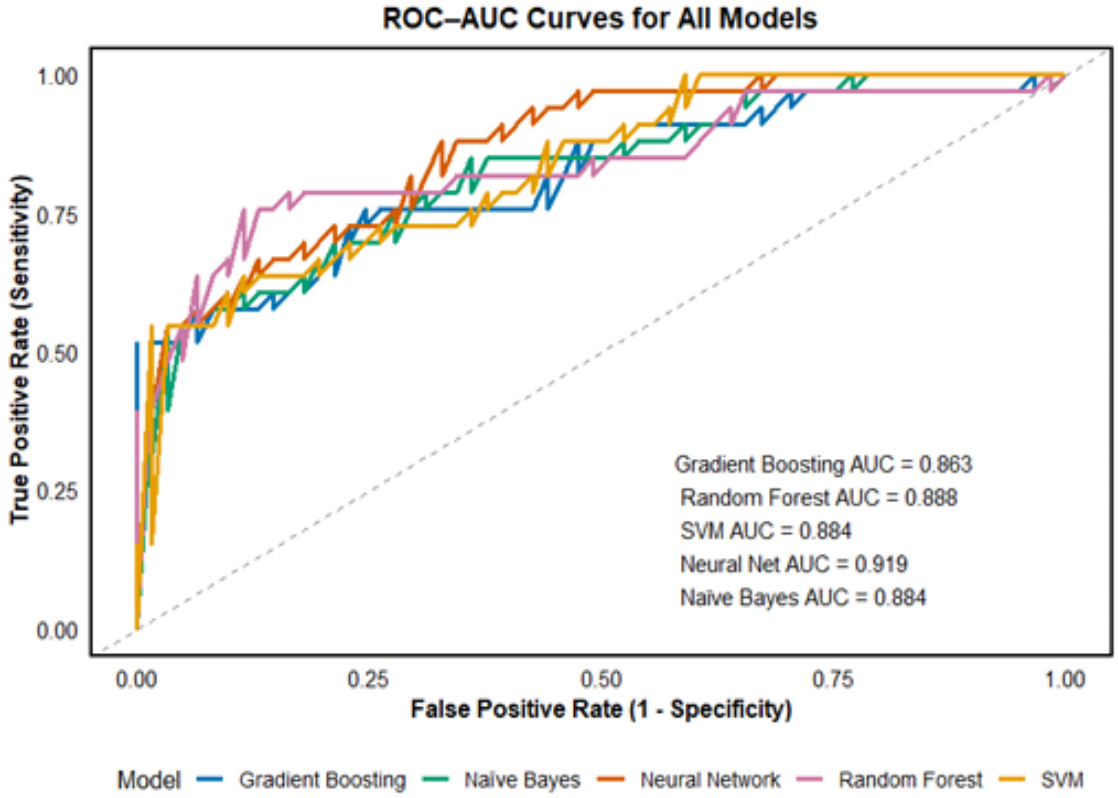


Figure 4.14: ROC–AUC analysis

4.7.2 Confusion Matrices

Interpretations:

Due to the significant class imbalance in the dataset, even with SMOTE resampling, all models struggled to accurately predict and classify mild and severe dementia cases, resulting in a considerable number of misclassifications in the CDR 0.5 and CDR 1 classes. It is worth noting that the CDR 2 class was not represented in our test set.

- Neural Networks accurately identified most of the non-demented individuals (CDR0), successfully classifying 58 out of 61 participants. This was supported by the highest AUC score of 0.919.
- Random Forest demonstrated outstanding performance, accurately classifying 17 of the 33 very mild dementia cases and achieving an AUC score of 0.888.
- Naïve Bayes, Gradient Boosting, and Random Forest exhibited a competitive performance in classifying moderate dementia (CDR 1) cases, correctly predicting 5 out of 10 cases.

- Support Vector Machine, just like other models, exhibited a similar pattern of misclassification in the CDR 0.5 and CDR 1 classes.

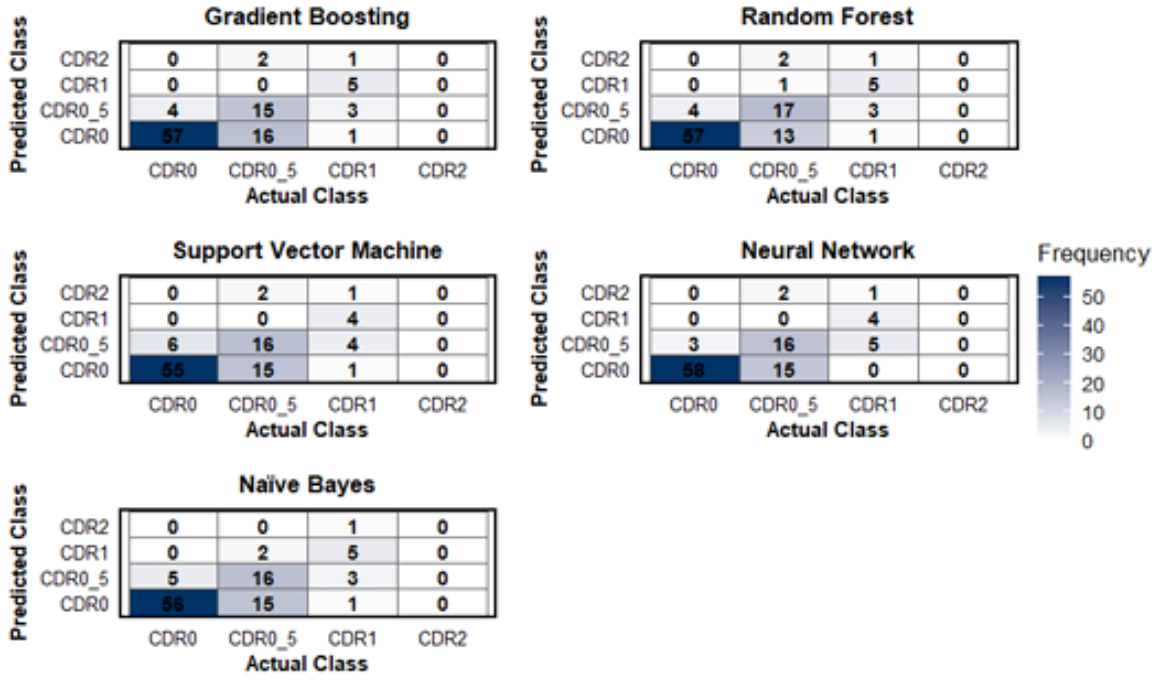


Figure 4.15: Combined confusion matrices for all the models

4.8 Model Selection

The Random Forest model was identified as the best classifier according to the evaluation metrics mentioned earlier, attaining an overall accuracy of 93.1% on the integrated dataset, comprising both the training and test sets. This demonstrates the model's exceptional performance. The significance of each predictor variable in the chosen Random Forest model was evaluated using the Gini index. The Gini index measures the extent to which a variable reduces impurity in a decision tree, indicating the significance of each predictor within the model.

The analysis revealed the third principal component as the most important predictor, with a 42.9% mean decrease in the Gini index. The second principal component also exhibited excellent predictor performance, with a 31.6% mean decrease in the Gini index. This indicates that alterations in brain structure, scores related to cognitive performance, and the clinical characteristics represented by these principal components play a significant role in the advancement of cognitive deterioration. The fifth principal component followed closely (26.3%), PC4 (23.8%), PC1 (21.7%), and Gender (5.2%).

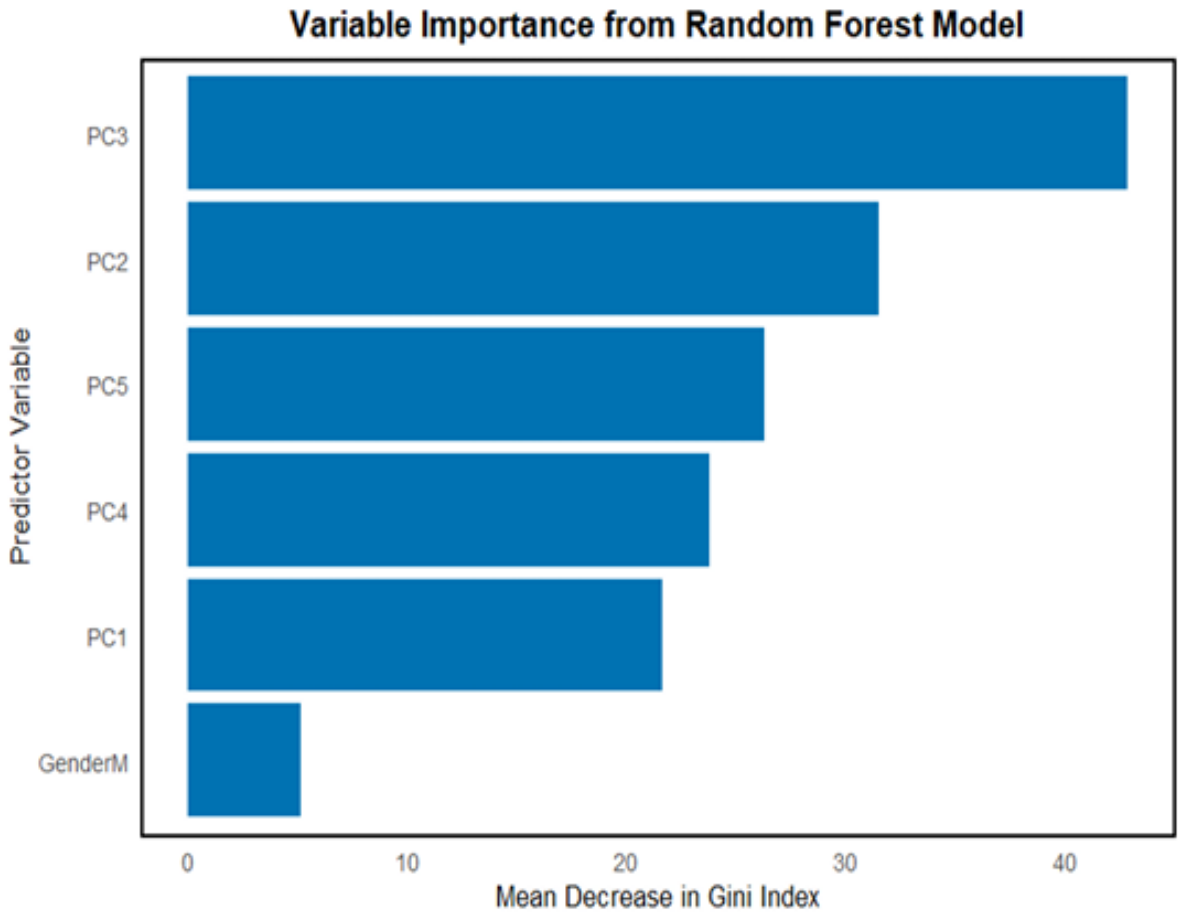


Figure 4.16: Variable importance analysis

4.9 Analysis in Connection to Prior Research

The results of this research closely align with the contributing factors for predicting dementia and/or Alzheimer's disease discussed in the Literature Review section. Alzheimer's disease arises from a complex interplay of clinical demographics, brain-related traits, and factors impacting cognitive function, rather than a single element. The predictive models developed in this study reflected these interactions and confirmed many associations identified in previous studies.

- **Cognitive Function scores**

  Previous research (Grundman et al., 2002; Mouton et al., 1998) has consistently demonstrated the importance of cognitive evaluation tools like the MMSE and CDR as effective indicators of dementia severity in individuals with mild cognitive impairment (MCI), functioning as a predictive measure for the potential progression to Alzheimer's disease (AD). This aligns closely with the analysis conducted in this study. The selected predictive model for Alzheimer's disease (AD) revealed the third principal component as the most important predictor, which is heavily loaded with the MMSE variable.

- **Brain-Related traits**

  The findings of this study support previous research on the importance of brain characteristics, including both normalized total brain volume and calculated intracranial volume. In agreement with the work of (Apostolova et al., 2012; Jack et al., 2000), reductions in brain size indicate the progression of Alzheimer's disease or variations in existing anatomy, as evidenced by the significance of the third and fifth principal components, which are highly correlated with nWBV and eTIV, respectively. Nonetheless, the inverse relationship observed between the severity of dementia and nWBV closely corresponds with previous research on the Atrophy-Cognition model, which suggests that major declines in critical brain regions, such as hippocampal atrophy, result in comparable cognitive symptoms.

- **Clinical Demographics**

  The research findings reaffirm previous literature on the role of clinical demographics in predicting and categorizing the progression of dementia. The impact of socioeconomic status and education level, as evidenced through the selected model's variable importance analysis highlighting the significance of the second principal component, and predominantly influenced by these two factors, further supports the cognitive reserve hypothesis introduced by Stern (2007), suggesting that individuals with higher educational attainment tend to experience a delayed onset or more gradual progression of dementia symptoms.

Furthermore, this study supports the research methods commonly applied in the prediction modeling of Alzheimer's disease. While conventional statistical techniques such as Multivariate Analysis of Variance, correlation, and regression analysis have played a crucial role in examining differences in brain structures between individuals with dementia and those without, as well as in associating cognitive scores with clinical and neuroimaging markers of cognitive decline (Cole et al., 1993; Liu-Seifert et al., 2016), they face constraints due to their assumptions of linearity, independence, and balanced class distributions. These constraints limit their effectiveness in capturing the intricate, nonlinear relationships present in actual clinical data, as evidenced in our findings from the longitudinal OASIS dataset utilized in this research.

The machine learning models utilized in this study, especially Random Forest and Gradient Boosting, demonstrated superior predictive accuracy compared to conventional methods. In line with the results from Genuer et al. (2010b) and Fabris et al. (2018), Random Forest exhibited the highest level of predictive accuracy, highlighting the algorithm's effectiveness in addressing imbalanced datasets in our study, thus lowering the risk of misclassifying subjects with severe dementia when compared to other machine learning classifiers used here. These findings are consistent with the work of Song et al. (2021), who showed that Random Forest predicts AD more reliably using fewer features.

It is essential to acknowledge that this study, in summary, adds to the existing literature by showing that machine learning techniques, specifically Random Forest and Gradient Boosting, can significantly enhance the predictive accuracy of models assessing the progression of Alzheimer's disease, especially in scenarios where conventional statistical methods struggle to address non-linear and complex relationships among demographic, clinical, and structural brain factors. Furthermore, the findings highlight the intricate, multifactorial aspects of Alzheimer's disease progression, which collectively influence the variations observed in dementia severity.

4.10 Discussion and Interpretation of Findings

The predictive models created in this research to forecast and classify the severity of dementia provided valuable insights into the factors contributing to the disease. The ensemble methods, especially Gradient Boosting and Random Forest, outperformed individual learning techniques such as Neural Networks, Support Vector Machines, and Naive Bayes, highlighting their ability to build a collection of classifiers that vote on predictions to enhance their forecasting capabilities. These results support existing research and address the complex non-linear interactions, interdependencies, and significant class imbalances that are typical in clinical and biomedical datasets. The relatively poor performance of these specific learning algorithms highlights the constraints noted in the

existing literature. Their challenges in managing non-linearities and imbalanced datasets, such as those faced by neural networks and naive Bayes, along with the meticulous kernel adjustments needed for SVMs, often result in interpretative difficulties.

The analysis of the research indicated once more that various factors, such as age, normalized whole-brain volume, estimated total intracranial volume, level of education, individual socioeconomic status, and Mini-Mental State Examination scores, play a crucial role in precisely modeling the progression of Alzheimer's disease. The analysis of variable importance underscored these factors due to their strong correlation with the principal components retained for modeling disease severity. Gender had a partial influence during the predictive modeling process; hence, sex-based differences might not significantly impact progression when considering clinical and imaging characteristics. This finding somewhat contradicts existing literature, as women are generally more prone to developing dementia, primarily attributed to their higher life expectancy in old age (Andrew and Tierney, 2018).

The failure of the K-means clustering technique to yield clear, distinct, and non-overlapping subgroups supports the notion proposed in prior studies, suggesting that Alzheimer's disease is perceived as a progressive condition ranging from asymptomatic preclinical stages to complete dementia, rather than a series of separate clinical phases. This observation, however, contrasts with the staging models presented by Butler Hospital and the Alzheimer's Association, which categorize the disease into clinical stages considered important for advancing effective disease-modifying therapies (DMTs) (Butler Hospital, nd; Alzheimer's Association, 2025).

To summarize, the use of preprocessing methods, such as the Synthetic Minority Oversampling Technique and Principal Component Analysis, effectively addressed the significant class imbalance and streamlined the dataset's dimensions, thereby enhancing its interpretability. The integration of these preprocessing strategies with ensemble learning models presents a coherent approach for predicting the advancement of dementia. The results of this research highlight the significant contributions that machine learning techniques, particularly Random Forest and Gradient Boosting, make to modeling the progression of Alzheimer's disease within the specified study framework. Nonetheless, they also emphasize the necessity for hybrid models that combine the interpretability of traditional statistical methods with the predictive capabilities of machine learning.

CHAPTER FIVE
Summary of Findings, Conclusion, and Recommendations

5.0 Overview

This chapter presents the summary of findings, conclusions, and recommendations arising from the study on the application of statistical and machine learning approaches to exploring brain structural changes in Alzheimer's disease progression. It highlights the key findings on the factors associated with dementia severity and evaluates the predictive performance of the models developed. The chapter further draws conclusions from the results, provides recommendations for the practical application of the findings, discusses the limitations encountered in the study, and identifies areas for future research.

5.1 Summary of Findings

The primary objective of this research was to investigate the gradual changes in brain structure and cognitive impairment associated with Alzheimer's disease. Specifically, the study utilized a combination of statistical techniques and machine learning algorithms to predict and categorize the severity of the disease, while also assessing the effectiveness of these approaches.

The analysis indicated that the dominant factors contributing to Alzheimer's disease severity included Age, Socioeconomic Status, MMSE, normalized Whole Brain Volume (nWBV), and estimated Total Intracranial Volume (eTIV). The Variable Importance Analysis highlighted the predictive significance of the retained principal components and their relationships with these variables. Consistent with the existing literature discussed in Chapter Two, these findings demonstrate the importance of cognitive function scores, global brain features, and clinical demographics in analyzing dementia severity. Gender, however, had relatively low variable importance in the predictive analysis. This suggests

that, within the dataset and modeling framework used in this study, gender contributed less to the prediction of dementia severity than the other variables. This finding should be interpreted cautiously, particularly in view of the gender imbalance in the dataset and the possibility that the observed pattern may not fully reflect gender-related differences in Alzheimer's disease progression.

Additionally, the results from the comparative modeling indicated notable variations in predictive performance between the ensemble methods and the standalone learning techniques across the five predictive models. The standalone learning techniques—Support Vector Machines, Neural Networks, and Naive Bayes—exhibited relatively lower overall accuracy, ranging from 73.6% to 77.7%. Their AUC scores also varied, with the Neural Network achieving the highest discriminative capability among all the models. Although all models faced challenges with misclassification due to significant class imbalance, even after applying the SMOTE resampling technique, the ensemble methods—Random Forest and Gradient Boosting—achieved the highest overall accuracy on the combined dataset, recording 93.1% and 92.6%, respectively. However, the Neural Network recorded the highest AUC of 91.9%, indicating its strong ability to distinguish between dementia severity classes. Random Forest followed with an AUC of 88.8%, while Gradient Boosting recorded an AUC of 86.3%. These findings demonstrate that predictive performance varied depending on the evaluation metric considered, with Random Forest providing the highest overall accuracy and the Neural Network achieving the strongest discriminative performance.

The key discoveries indicate that integrating neuroimaging with clinical information enhances the capability to predict and categorize the severity of dementia, providing a useful framework for assessing Alzheimer's disease progression and supporting ongoing disease assessment.

5.2 Conclusion

The findings of this research underscore the significance of cognitive evaluations, brain volume measurements assessed through magnetic resonance imaging (MRI), and clinical variables such as socioeconomic status, age, and gender in the assessment of Alzheimer's disease progression and the classification of dementia severity. These findings support previous studies that emphasize the importance of cognitive, demographic, and structural brain characteristics in understanding dementia severity. The knowledge acquired from this study is expected to contribute to further research in clinical practice and neuroscience, particularly in approaches aimed at improving the assessment and monitoring of Alzheimer's disease. Understanding the progression of Alzheimer's disease, alongside classifying and anticipating the severity of dementia, is crucial for identifying individu-

als who may require closer monitoring and appropriate interventions. This foundation can support additional research focused on identifying factors associated with disease progression and improving the prioritization of individuals at greater risk.

Among the models evaluated, the ensemble learning techniques, particularly Random Forest and Gradient Boosting, demonstrated the strongest overall classification accuracy. Random Forest achieved an overall accuracy of 93.1%, while Gradient Boosting recorded 92.6%. However, the Neural Network achieved the highest AUC of 91.9%, demonstrating that the best-performing model depended on the evaluation metric considered. The standalone learning techniques generally recorded lower overall accuracy, suggesting that they were less effective in capturing the complex relationships within the dataset. Although Principal Component Analysis helped reduce the dimensionality of the dataset and address multicollinearity, challenges associated with class imbalance and the potential for overfitting remained important considerations.

Overall, the results highlight the capacity of predictive modeling as a valuable decision-support approach for assessing dementia severity and identifying individuals who may require further clinical evaluation. With additional improvements in class representation, model validation, and the incorporation of larger and more comprehensive datasets, machine learning techniques such as Random Forest and Gradient Boosting could provide useful support for healthcare professionals and policymakers in developing more effective approaches to the assessment and monitoring of Alzheimer's disease.

5.3 Recommendations

Based on the findings and conclusions of this study:

- It is essential to prioritize the potential clinical application of ensemble-based predictive models. Healthcare facilities and clinical researchers should consider utilizing ensemble classification techniques, such as Random Forest and Gradient Boosting, alongside appropriate clinical assessments when developing predictive decision-support systems. These models could support the identification of individuals at elevated risk of neurodegeneration and contribute to early intervention and monitoring efforts.
- Enhancing data quality and ensuring adequate class representation are essential. Institutions that collect data on dementia and Alzheimer's disease should work towards improving the representation of different dementia severity categories to enhance the prediction and classification of disease severity while minimizing reliance on SMOTE resampling. This is particularly important for severe dementia cases, which were underrepresented in this research. Including additional variables

in the dataset, such as genetic and environmental factors, relevant biomarkers, the Global Deterioration Scale (GDS), and behavioral and lifestyle factors, could further improve model performance.

- Well-designed surveillance and data collection systems should be established at both national and global levels to monitor disease prevalence and associated risk factors. Such systems would provide more comprehensive information on the burden and progression of Alzheimer's disease and support the evaluation of strategies aimed at reducing its impact.

5.4 Limitations of the Study

Although this study met its objectives, it was subject to certain limitations. The dataset utilized was obtained from a publicly available source and may not fully represent the complexities of dementia in real-life clinical settings. Consequently, the results may vary if the models are applied to other datasets, such as the Alzheimer's Disease Neuroimaging Initiative (ADNI) database. Furthermore, class imbalance posed a challenge because severe dementia cases were less represented, complicating the models' ability to predict high-severity outcomes effectively. Although SMOTE was applied to address this imbalance, it may not completely resolve the difficulties associated with predicting underrepresented classes. Lastly, although this study examined five modeling techniques, other advanced algorithms and hybrid approaches could be investigated in future research to improve predictive performance.

5.5 Suggestion for Future Research

Drawing from the current findings, subsequent research should aim to apply the models developed in this study to Alzheimer's disease data sourced from both primary and secondary databases. This will provide more robust validation of the models' predictive capabilities and practical significance. Additionally, it is essential to use larger and more diverse datasets that feature participants from different ethnicities, education levels, and socioeconomic backgrounds to improve the models' generalizability and ensure that the predictive frameworks are relevant across various populations and clinical settings. Furthermore, future research should extend beyond prediction to assess the impact of actual policy and clinical interventions in mitigating cognitive decline.

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